

Managerial Economics

Premium Study E-Book & Complete Exam Guide

Concepts • Intuition • Derivations • PYQs

IIT Madras — BS in Data Science & Applications

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About This E-Book

This document is a **premium textbook-quality study guide** for the IIT Madras course *Managerial Economics* (Prof. Vimal Kumar). It synthesises every concept from Weeks 1–4 of the lectures and transcripts, expands every bullet point into a full beginner-friendly explanation, provides hundreds of worked examples and exam tips, and solves **every PYQ from the last 5 terms** with deep reasoning. It is designed to be **self-sufficient**: reading this file alone is enough to crack the quizzes and end-semester examinations.

How to use: read each Week sequentially; do the practice questions; then study the PYQ section to lock in patterns and traps.

“Economics is a science of decision-making” — Prof. Vimal Kumar

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Part I — Complete Weekly Notes

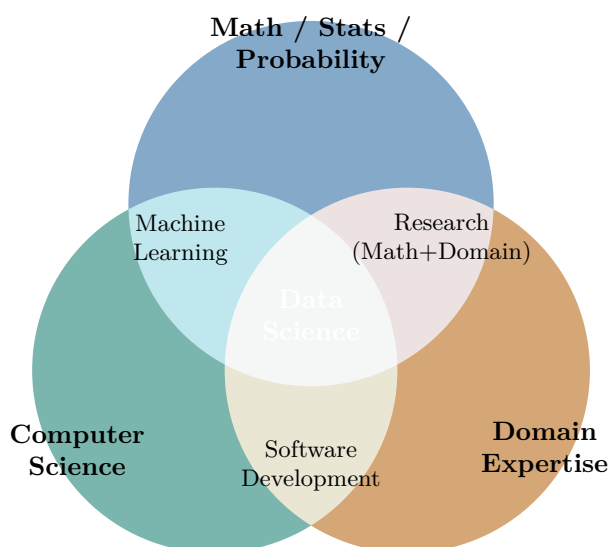
1. Week 1 — Foundations of Managerial Economics

1.1 Why Study Economics? — The Big Picture

Intuition

You are studying **Managerial Economics** as part of a Data Science programme. Why? Because a great data scientist needs *three* things working together: mathematics/statistics, computer science, *and* **domain expertise**. Economics is arguably the most powerful domain for a data scientist because it is itself quantitative, social, and rich in causal reasoning.

Prof. Kumar opens the course with a **Venn diagram** of three circles — mathematics & statistics, computer science, and domain expertise. Where any two intersect, you get a specialised field; where all three meet, you get **data science**.



1.1.1 What is Economics? — Multiple Lenses

Three complementary definitions to internalise:

1. **A science of decision-making.** Every moment we choose between alternatives; economics studies the *logic* of those choices.
2. **(Marshall, 1885)** “A study of human behaviour in the *ordinary business of life*.” — the emphasis on *ordinary* is critical: we model regular, day-to-day decisions, not eccentric ones.
3. **A study of allocating scarce resources to satisfy individual wants or desires.** The four keywords are: *allocating, scarce, individual, wants/desires*.

Core Definition — Economics

Economics is the framework for studying how decision-making units (individuals, households, firms, governments) allocate **scarce** resources among **unlimited wants and desires**. Mathematically, it is an **applied course in constrained optimisation**.

Why Scarcity Matters

If resources were unlimited, no choice would be necessary — everyone could have everything. The reason economics exists is the unavoidable gap between *unlimited wants* and *limited means* (money, time, raw materials, attention). Even Mukesh Ambani has a scarce resource: *time*. Therefore even the world’s richest person makes economic choices every day.

1.1.2 What is Managerial Economics?

Definition — Manager and Managerial Economics

A **manager** is a person who directs resources to achieve a stated goal.

Managerial Economics is the study of how to direct scarce resources in the way that *most efficiently* achieves a managerial goal. Equivalently, it is the **application of economics (mainly microeconomics) to business problems**.

The three branches of Economics.

- **Microeconomics** — the study of individual decision-makers (a household, a firm, a consumer).
- **Macroeconomics** — the study of aggregate phenomena (inflation, unemployment, GDP, interest rates).
- **Econometrics** — the measurement of economic phenomena combining statistics with economic theory; especially *causal inference from observational data*.

Managerial Economics draws almost entirely from **microeconomics**.

The three players of microeconomics. **Producers / Firms** (sellers) — **Consumers / Buyers** — and the **Market** where they meet (real or virtual).

1.1.3 Why Econometrics is Special (Brief)

In classical statistics (e.g. physics experiments) you *control* X and observe Y — strong notions of independent vs. dependent variables. In economics you cannot “vary education” the way you vary the length of a pendulum: both X and Y are stochastic. This is why econometrics, focused on **causal inference from observational data**, is distinct from textbook statistics.

1.2 The Tools We Will Use

Remember This

Four mathematical/statistical tools dominate this course:

- (1) Constrained Optimisation, (2) Equilibrium Analysis,
(3) Comparative Statics, (4) Regression Analysis (lightly).

1.2.1 Variables, Parameters, and Models

A **model** is a simplified representation of reality. Like a metro map that ignores buildings to highlight stations, our economic models discard inessential details.

- **Exogenous variables** — given from outside the model (e.g. price of raw materials faced by a small firm).
- **Endogenous variables** — determined inside the model (e.g. how much raw material the firm decides to buy).
- **Parameters** — fixed features of the environment (depreciation rate, worker wages set by contract).

1.2.2 Tool 1 — Constrained Optimisation

The generic problem:

$$\max_x f(x) \quad \text{s.t.} \quad g(x) \leq c$$

where $f(x)$ is the **objective function** (what you maximise — e.g. profit or utility), x is your **choice variable**, and $g(x) \leq c$ is the **constraint** (limited resources). Minimisation problems (e.g. cost minimisation) are perfectly symmetric.

- **Consumers** maximise *utility* (level of satisfaction).
- **Managers / firms** maximise *profit* (or minimise *cost*).

1.2.3 Tool 2 — Equilibrium

Equilibrium is the **balance of opposing forces**: e.g. buyers wanting low prices vs. sellers wanting high prices. Price is the coordinating mechanism. In equilibrium, no agent has any *incentive* to change behaviour.

1.2.4 Tool 3 — Comparative Statics

If you take a “photograph” of an equilibrium, change an exogenous variable, then take a “photograph” of the new equilibrium, and *compare* them — that is comparative statics. You ignore the transition (which would be comparative *dynamics*, a macro topic).

Exam Tip

Comparative *statics* = compare two equilibrium photographs. Comparative *dynamics* = study the video between them. This course focuses on **statics only**.

1.2.5 Tool 4 — Regression Analysis

For demand estimation, the firm collects data and runs a regression of the form $Y = X\beta + \varepsilon$. Unlike a deterministic equation, the error term ε captures unobserved factors. We will only *briefly* touch regression.

1.3 The Apartment-Rental Problem — Building Demand from Scratch

Real-World Application

You own an apartment complex with **5 vacant units**. Ten potential tenants approach, each with a maximum willingness-to-pay (WTP, in thousand rupees per month):

Person	Max WTP (in Rs000's / month)
1	70
2	22
3	55
4	15
5	22
6	35
7	55
8	50
9	60
10	65

Question: What is the maximum (and minimum) rent at which all 5 apartments are rented?

1.3.1 Brute-Force Reasoning

Sort the WTPs in decreasing order:

70, 65, 60, 55, 55, 50, 35, 22, 22, 15.

A rent P attracts every tenant whose $WTP \geq P$.

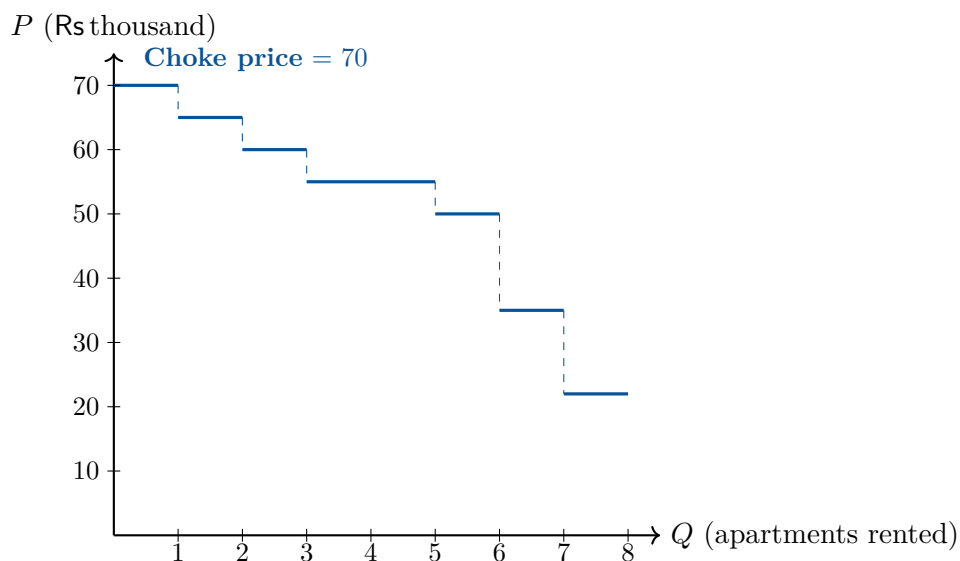
- $P > 70$: nobody rents.
- $65 < P \leq 70$: only person 1.
- $60 < P \leq 65$: persons 1, 10 (2 tenants).
- $55 < P \leq 60$: persons 1, 10, 9 (3 tenants).
- $50 < P \leq 55$: persons 1, 3, 7, 10 (4 tenants — two tied at 55).
- **$35 < P \leq 50$: persons 1, 3, 7, 8, 10 (5 tenants!) ✓**
- $22 < P \leq 35$: 6 tenants. And so on.

Remember This

Answer. To rent exactly 5 apartments, set rent $P \in (35, 50]$ thousand rupees. The *maximum* rent is Rs 50,000 and the next breakpoint downward is Rs 35,000.

1.3.2 Generalising to a Demand Function

For each price P , count the number of people whose $WTP \geq P$. That number is the **quantity demanded** $Q(P)$. Plotting P on the y-axis (a tradition since Marshall, 1885) and Q on the x-axis produces a **step-function** demand curve.



Definitions — Demand Vocabulary

- **Reservation price** of a buyer: the maximum amount he/she is willing to pay for **one additional unit**. (Equivalent here to “Max WTP.”)
- **Choke price**: the price above which no one buys anything. (Here: 70.)
- **Demand function** $Q(P)$: maps each price P to the quantity demanded.
- **Inverse demand function** $P(Q)$: maps each quantity Q to the price that would clear that quantity.

1.3.3 Approximating the Step Function by a Straight Line

If WTPs are dense, the steps shrink and the curve smooths out. A common smooth approximation is:

$$Q = \max\{30 - P, 0\}, \quad \text{equivalently} \quad P = 30 - Q \quad (\text{for } 0 \leq Q \leq 30).$$

Formula Insight

The $\max\{\cdot, 0\}$ wrapper is essential: without it, plugging $P = 40$ into $Q = 30 - P$ gives $Q = -10$, which is economically meaningless. Quantity cannot be negative.

Common Mistake

Many students confuse *demand* with *quantity demanded*.

- A change in **price** causes a *movement along* the same demand curve — this is a change in **quantity demanded**.
- A change in **any other factor** (income, taste, price of substitutes/complements) causes a *shift of the entire curve* — this is a change in **demand**.

1.4 Modelling the Buyer — Three Behavioural Assumptions

We treat each buyer as a **rational decision-maker**. Rationality here means **consistency** of choices, not perfection.

1. **Each buyer has a (subjective) reservation price r** for each unit.
2. **Buyers maximise consumer surplus $r - P$** when buying at price P .
3. **Among alternatives**, the buyer picks the option with the highest surplus.

Consumer Surplus

For a buyer with reservation price r purchasing at market price P :

$$\text{Consumer Surplus (CS)} = r - P.$$

This is the **gain** (in money-equivalent units) from the transaction. The buyer participates only if $\text{CS} \geq 0$ and chooses the alternative with the largest CS.

1.4.1 Why “Rational”? — A Defence

The assumption of rationality lets us *predict* behaviour. If an observed action looks irrational, we typically search for hidden context rather than abandoning the framework. Most decisions are consistent enough that the model is useful.

1.4.2 Reservation Price Diminishes with Consumption

If you’ve never had a mango this summer, your reservation price for the first one might be Rs 100. For the second mango that day, perhaps only Rs 60. Reservation price is *maximum WTP for one additional unit*, and the additional value usually falls — this is the seed of the **Law of Diminishing Marginal Utility** (Week 4).

1.5 Limitations of the Rationality Assumption

Common Mistake

Three classic violations students should know:

1. **Context-dependent reservation prices.** Same cold drink: you'd pay more for it fetched from a 5-star hotel than from a sack. The *good* is identical; the *price you accept* is not.
2. **Percentage vs. absolute thinking.** People drive 1 km to save Rs 1000 on an Rs 8000 Kindle (12.5%), but not on an Rs 80,000 iPad (1.25%) — though the absolute saving is identical.
3. **Reference prices instead of true CS maximisation.** Buyers often use a remembered “fair price” instead of comparing all options.

These are descriptive failures of rationality and form the basis of *behavioural economics* — beyond our syllabus, but useful to know.

1.6 Modelling the Seller — The Profit Motive

1.6.1 Profit, Revenue, Cost

Formula Insight

Profit identity:

$$\pi = TR - TC, \quad TR = P \cdot Q, \quad TC = C(Q).$$

- π (Greek pi) = profit.
- TR = total revenue = price per unit $\times$ quantity sold.
- TC = total cost, a function of Q (more output usually $\Rightarrow$ more cost).

Two ways to grow profit. The seller can either (a) raise price (**higher margin**) but typically sell less, or (b) lower price (**higher volume**) but earn less per unit. The right mix depends on the *market environment*.

1.6.2 Items that Affect Per-Unit Price

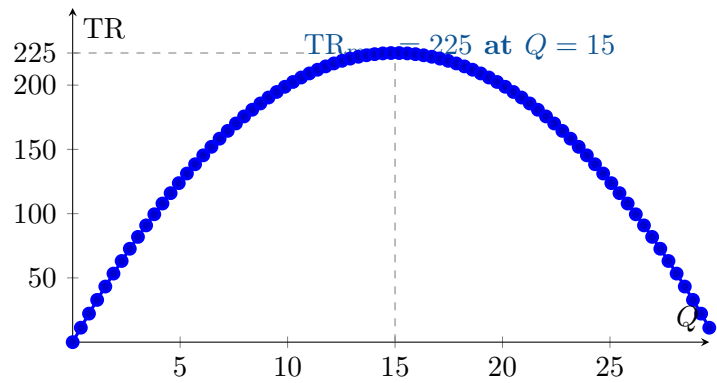
- **Discount** — granted up-front for early payment (e.g. “1/10, net 30” = 1% off if paid within 10 days; full amount due in 30).
- **Rebate** — given *after* purchase if the buyer completes paperwork. The psychological idea: many buyers never claim.
- **Returns** — newspaper vendors can return unsold copies for partial/full credit.
- **Allowance** — supermarket: “Display my product prominently and I’ll discount the price.”

1.7 The Revenue Function and the Inverted-U

With linear inverse demand $P = 30 - Q$, total revenue is

$$TR(Q) = P \cdot Q = (30 - Q)Q = 30Q - Q^2.$$

This is a downward-opening parabola. Setting $TR'(Q) = 30 - 2Q = 0$ gives the revenue-maximising quantity $Q^* = 15$, with maximum revenue $TR_{\max} = 15 \cdot 15 = 225$.



Intuition

The Inverted-U shape is a fundamental pattern: at very low Q , revenue is small because few units are sold; at very high Q , price has been driven down so much that revenue collapses. Maximum revenue lies somewhere in the middle.

1.8 Market Structures — A Preview

Structure	Setting	Pricing power
Perfect competition	Many sellers, many buyers, identical product	None — each firm is a <i>price taker</i> ; price = marginal cost.
Monopoly	One seller, many buyers	Maximum — still constrained by the demand curve.
Monopsony	Many sellers, one buyer	Maximum on the buyer’s side.
Oligopoly	A few sellers, many buyers	Strategic: each seller anticipates others (Game Theory!).

Remember This

In **perfect competition** firms keep undercutting each other until $P = MC$. Below MC , the next unit loses money, so no one drops the price further.

1.9 Pricing Techniques — The Big Picture

Every transaction creates economic value equal to (reservation price)–(marginal cost). The *size of the pie* is fixed by these two; the *division of the pie* depends on bargaining power.

Common techniques.

1. **Negotiation** (haggling): buyer and seller bargain.
2. **Take-it-or-leave-it** (fixed price): the seller pre-commits to a price.
3. **Auction**: the seller reveals buyers’ valuations.
4. **Price discrimination**: same product, different prices to different buyers (hard-cover vs. paperback; advance vs. same-day airline tickets).

1.10 The Sonu Nigam vs. Shaan Concert Problem — Opportunity Cost in Action

Real-World Application

Story. Yesterday you were given a free ticket to a Sonu Nigam concert (market price Rs 7500, resale value Rs 5000). Today you learn that Shaan is giving a concert the same evening; his ticket is also Rs 7500. If you had known yesterday, you would have bought the Shaan ticket and skipped Sonu.

Question. Should you attend Shaan's concert and not Sonu's?

1.10.1 Frame Both Choices as a Surplus Comparison

Let B_{SN} , B_{SG} denote your perceived benefit (in rupees) of watching Sonu Nigam and Shaan respectively. The original information tells us

$$B_{SG} - 7500 > B_{SN} - 7500 \iff B_{SG} > B_{SN}.$$

Good — but not yet conclusive. The free Sonu ticket now changes the cost structure:

- Net benefit of **going to Sonu Nigam**: $B_{SN} - 0 = B_{SN}$ (ticket already free).
- Net benefit of **going to Shaan**: $B_{SG} - (7500 - 5000) = B_{SG} - 2500$, because you can resell the Sonu ticket for Rs 5000 and apply that toward the Shaan ticket.

You should attend Shaan iff $B_{SG} - 2500 > B_{SN}$, i.e. iff $B_{SG} - B_{SN} > 2500$.

Remember This

Answer (in words). *It is not necessarily true.* You should attend Shaan only if your perceived extra benefit of Shaan over Sonu exceeds **Rs 2500** — the *opportunity cost* of giving up the resale value of the Sonu ticket.

The Bigger Lesson — Opportunity Cost

The **opportunity cost** of an action is the value of the next-best foregone option. Going to Shaan “costs” you the Rs 5000 you could have got by reselling Sonu's ticket. We will see this idea formalised in Week 2 as the distinction between **explicit** and **implicit** costs.

1.11 Generalised Decision-Making Framework

For *any* decision-maker (buyer or seller), let x_i be alternative i with benefit $b(x_i)$ and cost $c(x_i)$. Choose

$$x^* = \arg \max_i [b(x_i) - c(x_i)].$$

- For a **buyer**, b is perceived value (in money-equivalent terms) and c is price plus any opportunity cost.
- For a **seller**, b is total revenue and c is total cost.

Exam Tip

Why economists convert happiness to money. A buyer's “happiness” is subjective, but we can ask: *What is the maximum money you'd give up to avoid (or enjoy) X?* That amount is the money-equivalent of the experience. (Hypothetical: would you pay Rs 5000 to avoid a month in jail? Rs 1 lakh? Some price reveals your true disutility.)

1.12 End-of-Week Summary — Week 1

Week 1 Quick-Revision Sheet

Big Ideas:

- Economics = science of decision-making = constrained optimisation.
- Microeconomics studies producers, consumers, and the markets where they meet.
- Four mathematical tools: optimisation, equilibrium, comparative statics, regression.

Buyer Side:

- Reservation price $r = \max$ WTP for one additional unit.
- Consumer surplus $CS = r - P$. Buyer chooses option with highest CS.
- Demand curve is downward-sloping; choke price is where it hits the y-axis.

Seller Side:

- Profit $\pi = TR - TC$, $TR = P \cdot Q$.
- With linear demand $P = a - bQ$, TR is an inverted parabola.
- Revenue is maximised where $TR'(Q) = 0$; for $TR = aQ - bQ^2$, $Q^* = a/(2b)$.

Market Structures: Perfect competition (price-taker), Monopoly, Monopsony, Oligopoly.

Key Insight: Opportunity cost = value of the next-best forgone option.

1.12.1 High-Probability Quiz Points

- *Definition* of managerial economics, opportunity cost, reservation price, consumer surplus, choke price.
- *Movement along vs. shift of* a demand curve.
- Compute consumer surplus given r and P .
- Identify rent ranges (apartment-style table) at which n buyers participate.
- Decide whether to attend an event when a sunk-cost (free ticket) is involved.
- Find Q^* maximising TR for a linear demand curve.

1.12.2 Practice Questions — Week 1

1. Given the WTPs $\{80, 70, 65, 50, 30, 20\}$ for 6 buyers and 3 vacant apartments, find the range of rents at which all 3 are rented.
2. For inverse demand $P = 40 - 2Q$, compute the revenue-maximising Q^* and $TR_{\max}$.
3. State whether each is a change in *quantity demanded* or *demand*: (a) price of petrol rises; (b) a new study makes coffee fashionable; (c) consumers' income doubles.
4. A buyer's reservation price is Rs 800 and the market price is Rs 950. What is her consumer surplus? Does she buy?
5. You have a Rs 2000 non-refundable concert ticket. Tomorrow night, a friend offers you free tickets to a different concert worth Rs 3000 to you. Which do you attend? Why? (Hint: the Rs 2000 is sunk.)

2. Week 2 — Costs, Production, and Profit Maximisation

2.1 Motivating Problem — Drive to Delhi or Take the Train?

Real-World Application

You live in Kanpur (≈ 500 km from Delhi) and you are indifferent between driving and taking the train except for **cost**. The train ticket costs Rs 1000. A rental agency gives you the following yearly cost breakdown for your car, assuming an annual usage of 20,000 km:

Item	Annual (Rs)	Per km (Rs)
Insurance	10,000	0.50
Interest payment	20,000	1.00
Fuel & oil	10,000	0.50
Maintenance	10,000	0.50
Total	50,000	2.50

The naïve per-km total is Rs 2.50, so a 500-km trip would cost $500 \times 2.50 = \text{Rs } 1250$ — making the train (Rs 1000) appear cheaper.

Quiz Trap

This naïve answer is wrong! The mistake is treating every cost as “variable” when many are **fixed** or **sunk**. Read the rest of this chapter; you will revisit this problem in Week 4.

2.2 Cost Concepts — Explicit, Implicit, Sunk

Key Cost Definitions

- **Explicit cost** — direct monetary payment made by the firm (wages, rent, electricity bill, money paid to grocers).
- **Implicit cost** (*opportunity cost*) — the value of resources *owned* by the firm and used in the business, foregone from their next-best use (e.g. the salary you give up to run your own business; interest you’d earn on capital you invest yourself).
- **Sunk cost** — cost *already incurred and unrecoverable*, regardless of future actions.
- **Opportunity cost** of an action — the value of the next-best alternative foregone by undertaking it. *Includes both explicit and implicit costs.*

2.2.1 Accounting Profit vs. Economic Profit

	Accounting Profit	Economic Profit
Definition	TR – explicit costs	TR – explicit – implicit costs
Includes opportunity cost?	No	Yes
Used by	Accountants, tax authorities	Economists, decision-makers
Decision relevance	Limited	Direct

Why “Economic Profit” is the Right Concept

Suppose you run a shop with *accounting profit* of Rs 8 lakh/year. If you could earn Rs 10 lakh/year as a salaried employee, your *implicit cost* is Rs 10 lakh, and your *economic profit* is –Rs 2 lakh. You would be better off closing the shop and taking the job. Accounting profit hides this fact.

2.2.2 The Two Cardinal Lessons of Cost Analysis

Remember This

- (1) Consider implicit (opportunity) cost in every decision.
- (2) Ignore sunk costs — they are irrelevant to future decisions.

Example — The Fritos Problem (PYQ classic). You paid Rs 100 for 5 cases of Fritos that the customer rejected. Grocer will refund Rs 0.25 per dollar (i.e. Rs 25 total) on return. A fraternity offers Rs 55 for the lot.

- **Sunk cost** of the original purchase: Rs 100 (irrelevant!).
- **Opportunity cost** of selling to the fraternity = next-best foregone = Rs 25 (return to grocer).
- **Net gain** from frat sale = $55 - 25 = +\text{Rs } 30 > 0$. **Sell!**

2.3 Fixed Cost, Variable Cost, Average Cost

Decomposition of Total Cost

$$C(Q) = \text{TFC} + \text{TVC}(Q),$$

where:

- **Fixed cost** TFC — does *not* change with output Q (rent of factory, machinery purchased, manager's salary, insurance on a bus).
- **Variable cost** $\text{TVC}(Q)$ — changes with Q (raw materials, fuel, hourly labour).
- **Quasi-fixed cost** — almost fixed; jumps from 0 to a constant once production starts (e.g. startup cost of a hydro generator). We treat it as fixed in this course.

2.3.1 Average Costs

$$\text{AC}(Q) = \frac{C(Q)}{Q} = \frac{\text{TFC}}{Q} + \frac{\text{TVC}(Q)}{Q} = \text{AFC}(Q) + \text{AVC}(Q).$$

Common Mistake

Confusion alert:

- Fixed *total* cost is fixed; but **average fixed cost falls with Q** because the same FC is spread over more units. $\text{AFC}(Q) = \text{TFC}/Q$ is a hyperbola.
- Variable *total* cost rises with Q , but **average variable cost may rise, fall, or stay constant** depending on technology.

Quick numerical example. $\text{TFC} = 500$, $\text{TVC} = 5Q$ (constant marginal cost).

$$\text{AFC} = \frac{500}{Q}, \quad \text{AVC} = \frac{5Q}{Q} = 5, \quad \text{AC} = 5 + \frac{500}{Q}.$$

At $Q = 10$: $\text{AFC} = 50$, $\text{AVC} = 5$, $\text{AC} = 55$. At $Q = 100$: $\text{AFC} = 5$, $\text{AVC} = 5$, $\text{AC} = 10$.

2.4 Marginal Cost — The Cost of One More Unit

Marginal Cost — Three Equivalent Definitions

Rudimentary:

$$MC(Q) = C(Q + 1) - C(Q).$$

Discrete (allows backward step):

$$MC(Q) = \frac{C(Q + h) - C(Q)}{h} \text{ for } h = \pm 1.$$

Calculus (most accurate, for continuous Q): $MC(Q) = \lim_{h \rightarrow 0} \frac{C(Q + h) - C(Q)}{h} = \frac{dC}{dQ}.$

Marginal Cost Equals Marginal Variable Cost

Since $C(Q) = \text{TFC} + \text{TVC}(Q)$ and TFC is constant,

$$MC(Q) = \frac{dC}{dQ} = \frac{d \text{TVC}}{dQ}.$$

Fixed costs do *not* affect marginal cost.

2.4.1 The MC–AC Relationship (Crucial!)

Numerical intuition. If 10 units cost Rs 100 and the 11th unit costs Rs 11 (i.e. $MC = 11$), the new total is 111 over 11 units, so $AC = 10.09$. The MC (11) was higher than the old AC (10), and AC rose.

Calculus proof. Let $AC(Q) = C(Q)/Q$. Differentiate:

$$\begin{aligned} \frac{d AC}{dQ} &= \frac{d}{dQ} \left[\frac{C(Q)}{Q} \right] = \frac{C'(Q) \cdot Q - C(Q) \cdot 1}{Q^2} \\ &= \frac{1}{Q} \left[C'(Q) - \frac{C(Q)}{Q} \right] = \frac{1}{Q} [MC(Q) - AC(Q)]. \end{aligned}$$

The MC–AC Rule

- If $MC > AC \Rightarrow AC$ is *rising*.
- If $MC < AC \Rightarrow AC$ is *falling*.
- If $MC = AC \Rightarrow AC$ is *at its minimum* (or maximum).

Geometric fact: The MC curve cuts the AC curve at its minimum.

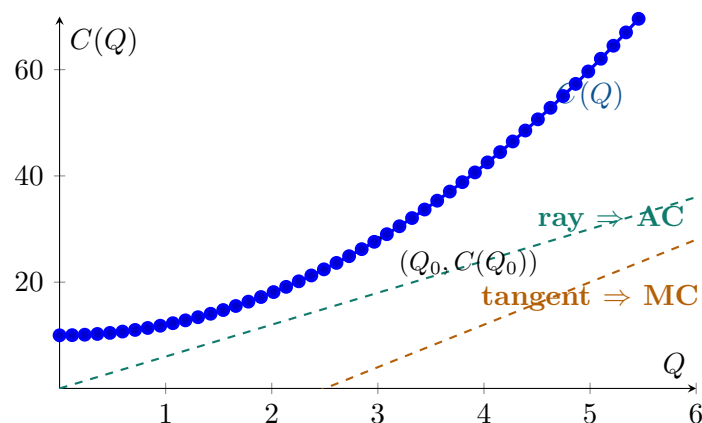
Cricket Analogy

Think of batting average. If you bat *above* your average today (your “marginal” score), your average rises. If you bat *below*, it falls. They are equal at the exact moment your average peaks or troughs.

2.4.2 Graphical Interpretation of MC and AC

On the total cost curve $C(Q)$ (with Q on the x-axis, C on the y-axis):

- AC at a point Q_0 = slope of the *ray* from the origin to the point on the curve.
- MC at a point Q_0 = slope of the *tangent* to the curve at that point.



2.5 Production Functions — The Firm as a Black Box

Definition

A **production function** maps inputs to maximum feasible output:

$$Y = f(X_1, X_2, \dots, X_n)$$

where Y is output (often labelled Q) and X_i 's are inputs. We typically use two inputs: **Labour** L and **Capital** K , writing $Y = f(L, K)$.

The letter K for capital is historical — from Marx's *Das Kapital* (German).

2.5.1 Two Concepts of Efficiency

Remember This

Technical (technological) efficiency: producing the maximum output from a given bundle of inputs — an *engineering* concept.

Economic efficiency: producing a target output at the *minimum cost* given input prices — a *managerial* concept.

The production function f already embeds technical efficiency. Economic efficiency requires us to also know input prices.

2.6 Three Canonical Production Functions

2.6.1 (1) Leontief / Fixed-Proportion

Inputs must be combined in a fixed ratio (sandwich example: 2 slices of bread + 1 spoon butter per sandwich).

$$Y = \min \left\{ \frac{X_1}{a}, \frac{X_2}{b} \right\}$$

With 5 bread slices and 2 spoons butter (and $a = 2, b = 1$): $Y = \min\{5/2, 2/1\} = \min\{2.5, 2\} = 2$ sandwiches. One slice of bread is wasted.

Isoquant. L-shaped (right angles). All technical-efficiency lies at the corner.

2.6.2 (2) Linear / Perfect Substitutes

Inputs are perfectly substitutable in a fixed exchange ratio:

$$Y = aX_1 + bX_2$$

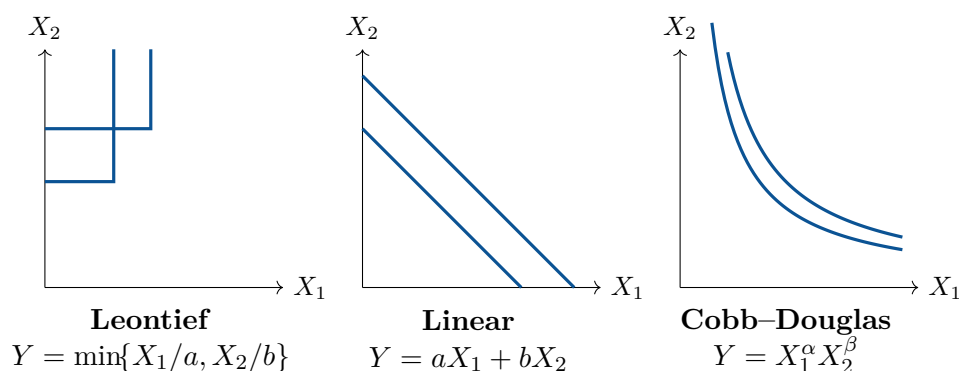
Isoquant. A straight line with constant slope $-a/b$ (in X_1 - X_2 space).

2.6.3 (3) Cobb–Douglas

The workhorse production function:

$$Y = A X_1^\alpha X_2^\beta$$

Isoquant. A smooth, convex hyperbola-like curve. As $X_1 \rightarrow 0$, $X_2 \rightarrow \infty$ to maintain output (and vice versa). Inputs are **imperfect substitutes**.



Identifying Which Production Function a Question Uses

PYQ language cues:

- “Trade a units of one input for b units of the other and keep output constant” $\Rightarrow$ **Linear**.
- “Exactly two workers needed per machine” / “fixed proportion” $\Rightarrow$ **Leontief**.
- “ $Y = K^a L^b$ ” or anything with exponents on inputs $\Rightarrow$ **Cobb–Douglas**.

2.7 Isoquants

Definition

An **isoquant** is a curve in input space (e.g. L - K plane) that connects all input bundles yielding the same level of output — a “level curve” of the production function.

Example. $Y = X_1 X_2$.

- $Y = 6$: $X_1 X_2 = 6$ — rectangular hyperbola.
- $Y = 10$: another hyperbola further from origin.

The *isoquant map* consists of several such curves. Higher isoquants correspond to higher output.

2.8 Total, Average, and Marginal Product (TP, AP, MP)

Let $Y = f(X)$ be the output when input X is used (with other inputs held fixed).

Productivity Definitions

$$\begin{aligned} \text{Total product (TP)} &= Y = f(X) \\ \text{Average product (AP)} &= \frac{Y}{X} = \frac{f(X)}{X} \\ \text{Marginal product (MP)} &= \frac{dY}{dX} = f'(X) \quad (\text{approx. } f(X+1) - f(X)) \end{aligned}$$

2.8.1 Marginal Products with Two Inputs

$$MP_L = \frac{\partial Y}{\partial L}, \quad MP_K = \frac{\partial Y}{\partial K}.$$

Diminishing marginal productivity (assumed): $\frac{\partial MP_L}{\partial L} < 0$, i.e. $f_{LL} < 0$.

2.8.2 MP–AP Relationship (Mirrors MC–AC!)

By the exact same calculus as before:

$$\frac{d AP}{dX} = \frac{1}{X} [MP - AP].$$

Remember This

- $MP > AP \Rightarrow AP$ rising.
- $MP < AP \Rightarrow AP$ falling.
- $MP = AP \Rightarrow AP$ at its maximum.

This is the production-side twin of the cost-side MC–AC rule. **Both come up almost every quiz.**

2.9 Forms of Business Organisation (Brief)

A snapshot of the four canonical legal forms (relevant to PYQ MCQs):

	Sole Prop.	Partnership	LLP	Corporation
Owners	1 (or family)	2 or more	2 or more	many (shareholders)
Registration	not needed	varies	required	required
Liability	unlimited	unlimited (shared)	limited	limited
Taxation	once (income)	once (income)	once	twice (corp + income)
Life of firm	until owner	until partners	longer	perpetual
Example	local store, taxi	law firm	Sony, Nike	Pepsi, Pfizer, Microsoft, Ford

In India: add **One Person Company (2013)**, **Pvt Ltd**, **Public Ltd**, **Section 8 (non-profit)**, **Joint Hindu Family business**, and **joint ventures**.

2.10 Profit Maximisation — The Marginal Principle

The Marginal Optimality Condition

A firm choosing Q to maximise $\pi(Q) = TR(Q) - TC(Q)$ obtains the first-order condition

$$\frac{d\pi}{dQ} = 0 \iff \boxed{MR(Q^*) = MC(Q^*)}$$

where $MR = d TR / dQ$ and $MC = d TC / dQ$.

Why? If $MR > MC$, the next unit adds more revenue than cost — increase Q . If $MR < MC$, the last unit adds more cost than revenue — decrease Q . Equality is the break-even on the margin.

Discrete case. If Q can only change by integer steps, choose Q^* where

$$MR(Q^*) \geq MC(Q^*) \quad \text{and} \quad MR(Q^* + 1) < MC(Q^* + 1).$$

2.10.1 Detailed Worked Example — Train Operations

Real-World Application

You are a regional manager for Indian Railways and must decide the number of trains to run daily between cities A and B. *Daily cost per train = Rs 15 lakh.*

Trains n	TC (RsL)	Total Benefit (RsL)	Avg. Benefit	Marginal Benefit
0	0	0	—	—
1	15	30	30	30
2	30	48	24	18
3	45	60	20	12
4	60	64	16	4
5	75	66	13.2	2

Marginal Cost = MC = 15 lakh (each additional train costs 15 L).

Decision rule. Run the next train if $MB \geq MC$.

- Train 1: $MB = 30 \geq 15$? Yes. Run.
- Train 2: $MB = 18 \geq 15$? Yes. Run.
- Train 3: $MB = 12 < 15$. **Stop.**

Optimal number of trains = 2. Brute-force across all 6 options gives the same answer; the marginal approach is faster and generalises to continuous problems.

2.10.2 Worked Example — Continuous Profit Maximisation

A firm faces price $P = 66$ (price-taker) and total cost $C(Q) = \frac{Q^3}{3} - 10Q^2 + 102Q + (\text{fixed})$. Find the profit-maximising Q^* .

Step 1 — MR. $TR = 66Q \Rightarrow MR = 66$.

Step 2 — MC.

$$MC = \frac{dC}{dQ} = Q^2 - 20Q + 102.$$

Step 3 — Set MR=MC.

$$Q^2 - 20Q + 102 = 66 \implies Q^2 - 20Q + 36 = 0.$$

Use $(Q - 10)^2 = Q^2 - 20Q + 100$, so $Q^2 - 20Q + 36 = (Q - 10)^2 - 64 = 0$, giving $(Q - 10)^2 = 64$, i.e. $Q - 10 = \pm 8$. So $Q = 2$ or $Q = 18$.

Step 4 — Which root? The MC curve is a U-shape. At $Q = 2$, MC is on the *falling* branch (any small increase in Q lowers MC below 66, so π would rise). Hence $Q = 2$ is a local *minimum* of profit, not a maximum.

At $Q = 18$, MC is on the *rising* branch and crosses MR from below, so this is the profit-maximising quantity. $Q^* = 18$.

Second-Order Check

The full second-order condition is $\frac{d^2\pi}{dQ^2} < 0$, i.e. $\frac{d MC}{dQ} > 0$ at Q^* . **So pick the root where MC is rising.** Equivalently: pick the higher of the two roots when MR is a horizontal line and MC is U-shaped.

2.11 End-of-Week Summary — Week 2**Week 2 Quick-Revision Sheet****Costs:**

- Total cost $C(Q) = \text{TFC} + \text{TVC}(Q)$, where TFC is fixed and TVC varies with Q .
- Explicit (paid out) vs. Implicit (opportunity) vs. Sunk (gone forever).
- Economic Profit = TR – explicit cost – implicit cost.
- $\text{AFC} = \text{TFC} / Q$ (falls with Q). $\text{AVC} = \text{TVC} / Q$. $\text{AC} = \text{AFC} + \text{AVC}$.
- $\text{MC} = dC/dQ = d \text{TVC} / dQ$ (fixed cost vanishes!).

Production:

- Three canonical functions: Leontief (fixed-proportion), Linear (perfect substitutes), Cobb–Douglas.
- Isoquant = combinations of inputs giving same Y .
- AP, MP — their relationship mirrors AC, MC.

Optimisation Master Rule: $\text{MR} = \text{MC}$ at the profit-maximising Q^* (with second-order check: MC rising through MR).

2.11.1 High-Probability Quiz Points (Week 2)

- Identifying explicit, implicit, sunk costs in a story (PYQ favourite!).
- Computing AFC, AVC, MC at a given Q from a TC equation.
- Recognising Leontief/Linear/Cobb–Douglas from verbal cues.
- Stating the MC–AC relationship.
- Profit maximisation: $\text{MR} = \text{MC}$.

2.11.2 Practice Questions — Week 2

1. $\text{TC}(Q) = 200 + 50Q + Q^2$. Find TFC, AVC at $Q = 10$, AC at $Q = 10$, and MC at $Q = 10$.
2. A bakery owner gives up a Rs 50,000/month salary to run the bakery. Monthly explicit costs are Rs 1,20,000. Revenue is Rs 1,60,000.
 - (a) Accounting profit?
 - (b) Implicit cost?
 - (c) Economic profit?
3. Match each to a production function: “perfect substitutes”, “fixed ratio”, “ $Y = K^{0.5}L^{0.5}$ ”.
4. For $Y = 4K^{1/2} + 4L$, find MP_K and MP_L . State whether each is constant or declining.
5. If MC equals AC, what does it imply about AC at that Q ?

3. Week 3 — MRTS, Cost Minimisation, Returns to Scale, Supply

3.1 Recap and Roadmap

Week 3 builds the bridge from the production side to the supply curve. Key new ideas:

- **Marginal Rate of Technical Substitution (MRTS)** — the slope of an isoquant.
- **Cost minimisation** — choosing inputs to produce a target output cheapest.
- The **cost function** $C(Q)$ for Leontief, Linear, and Cobb–Douglas technologies.
- **Returns to scale** and **short run vs. long run**.
- Deriving the **supply function** from the marginal cost curve.

3.2 Marginal Rate of Technical Substitution (MRTS)

MRTS — Definition

The **MRTS of labour for capital** is the amount of capital the firm can give up per unit of additional labour while keeping output constant:

$$\text{MRTS}_{L,K} = - \frac{dK}{dL} \Big|_{Y=\text{const}}.$$

It is the (absolute value of the) slope of the isoquant at a point.

3.2.1 Deriving the MRTS Formula

Take the total differential of $Y = f(L, K)$ holding Y fixed:

$$dY = f_L dL + f_K dK = 0 \implies \frac{dK}{dL} \Big|_{Y=\text{const}} = - \frac{f_L}{f_K} = - \frac{\text{MP}_L}{\text{MP}_K}.$$

Hence

$$\text{MRTS}_{L,K} = \frac{\text{MP}_L}{\text{MP}_K}$$

(taking magnitudes; the geometric slope is negative).

Intuition

If you give up a unit of capital, you lose MP_K units of output. To recover them you need $\frac{\text{MP}_K}{\text{MP}_L}$ units of labour. Hence one unit of capital “exchanges” for $\text{MP}_K / \text{MP}_L$ units of labour — and one unit of labour exchanges for $\text{MP}_L / \text{MP}_K$ units of capital. Be careful about direction!

Worked numerical example. $Y = LK$ (Cobb–Douglas with exponents 1, 1). $\text{MP}_L = K$, $\text{MP}_K = L$. So $\text{MRTS}_{L,K} = K/L$.

- At $(L, K) = (3, 4)$: $\text{MRTS} = 4/3$. Giving up 1 unit of K requires $3/4$ unit of L (to keep Y constant). Equivalently, an extra unit of L frees up $4/3$ units of K .
- At $(L, K) = (2, 6)$: $\text{MRTS} = 6/2 = 3$. Higher capital intensity $\Rightarrow$ labour is “more powerful” on the margin.

Convexity of isoquants. Cobb–Douglas isoquants are *convex* (bowed toward origin). Moving rightward along an isoquant, L rises and K falls, so MRTS decreases — the **diminishing MRTS**: as you accumulate more L , each extra unit substitutes for less K .

3.2.2 MRTS for the Three Canonical Technologies

Technology	Form	MRTS
Linear (perfect substitutes)	$Y = aL + bK$	a/b (constant)
Leontief (fixed proportion)	$Y = \min\{L/a, K/b\}$	0 or ∞ (kinked)
Cobb–Douglas	$Y = AL^\alpha K^\beta$	$\frac{\alpha K}{\beta L}$ (diminishing)

Exam Tip

PYQ: “A firm can always trade 3 units of labour for 1 unit of capital and keep output constant.” The constant exchange ratio $\Rightarrow$ **Linear** production function.

3.3 Cost Minimisation — The Manager’s Problem

Definition

The cost-minimisation problem with input prices w (labour wage) and r (capital rental):

$$\min_{L,K} wL + rK \quad \text{subject to} \quad f(L, K) \geq Y_0.$$

Isocost line: $wL + rK = C$ — the locus of (L, K) bundles with the same expenditure. Slope $= -w/r$. As C increases, the isocost line shifts outward parallel to itself.

3.3.1 The Tangency Condition (Interior Solution)

At a cost-minimising interior point (L^*, K^*) , the isoquant is tangent to the isocost line:

$$\text{MRTS}_{L,K} = \frac{\text{MP}_L}{\text{MP}_K} = \frac{w}{r}$$

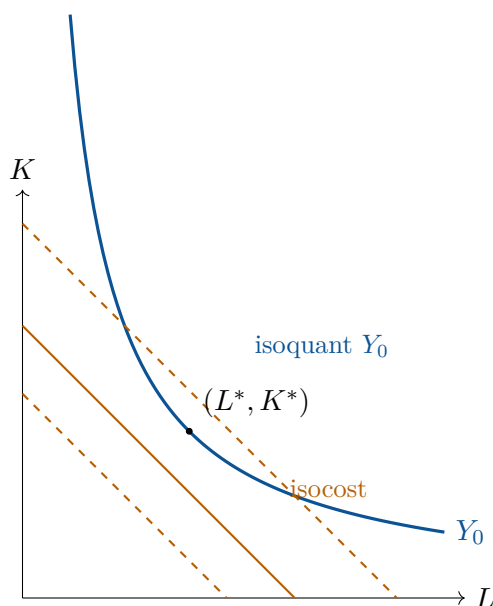
Equimarginal Principle

Rearranging the tangency condition:

$$\frac{\text{MP}_L}{w} = \frac{\text{MP}_K}{r}.$$

The marginal output **per rupee spent** should be equal across all inputs. If $\text{MP}_L/w > \text{MP}_K/r$, the firm gets “more bang per buck” from labour — it should use more labour and less capital, until the ratios equalise.

3.3.2 Visualising Cost Minimisation



3.3.3 Cost Minimisation for Leontief (Corner)

Production function $Y = \min\{L/a, K/b\}$ has L-shaped isoquants. At technical efficiency $L/a = K/b$, i.e. $L = aY_0$ and $K = bY_0$. The cost function is

$$C(Y_0) = w \cdot aY_0 + r \cdot bY_0 = (wa + rb) Y_0.$$

3.3.4 Cost Minimisation for Linear (Often Corner)

Production function $Y = aL + bK$. $MRTS = a/b$ is constant; isoquants are straight lines.

- If $w/r < a/b$ (labour is “cheap relative to its productivity ratio”), use **only labour**: $L = Y_0/a$, $K = 0$.
- If $w/r > a/b$, use **only capital**: $K = Y_0/b$, $L = 0$.
- If $w/r = a/b$, any combination works.

PYQ Recipe for Linear Cost-Minimisation

With $Y = aL + bK$, w , r :

1. Compute $MP_L/w = a/w$ and $MP_K/r = b/r$.
2. Use only the input with the higher ratio — it gives the most output per rupee.
3. Compute L^* or K^* from the constraint $aL^* = Y_0$ (or $bK^* = Y_0$).

Example (PYQ). $Y = 10L + 2K$, $w = 5$, $r = 2$, $Y_0 = 200$. $MP_L/w = 10/5 = 2$. $MP_K/r = 2/2 = 1$. Labour wins. So $L^* = 200/10 = 20$, $K^* = 0$. Cost = $5 \times 20 + 2 \times 0 = \100 .

3.3.5 Cost Minimisation for Cobb–Douglas (Interior)

$Y = AL^\alpha K^\beta$. $MRTS = \frac{\alpha K}{\beta L}$. Set $MRTS = w/r$:

$$\frac{\alpha K}{\beta L} = \frac{w}{r} \implies K = \frac{\beta w}{\alpha r} L.$$

Substitute into the constraint $Y_0 = AL^\alpha K^\beta$ and solve for L^* ; then K^* .

Concrete example. $Q = 50\sqrt{LK}$, $w = 5$, $r = 20$, $Q_0 = 1000$.

Here $\alpha = \beta = 1/2$. Tangency: $\frac{K}{L} = \frac{w}{r} = \frac{5}{20} = \frac{1}{4}$. So $K = L/4$.

Plug into $1000 = 50\sqrt{L \cdot L/4} = 50 \cdot L/2 = 25L$.

$$L^* = 40, \quad K^* = 10.$$

$$C^* = 5 \cdot 40 + 20 \cdot 10 = \$400.$$

3.4 The Total Cost Function

The cost-minimisation problem produces, as a by-product, the **long-run total cost function** $C(Q)$: the minimum total expenditure to produce each output level Q .

- **Leontief:** $C(Q) = (wa + rb)Q$ — linear in Q .
- **Linear (corner):** $C(Q) = \min\{w/a, r/b\} \cdot Q$ — linear in Q .
- **Cobb–Douglas** $Y = L^{1/2}K^{1/2}$, $w = 5$, $r = 20$: $C(Q) = 2\sqrt{wr} \cdot Q = 20Q$ (here $2\sqrt{5 \cdot 20} = 20$). Also linear!

In each of these three constant-returns cases, $C(Q)$ is linear and average cost is *constant* (more on returns to scale next).

3.5 Returns to Scale

Returns to Scale

For a production function f and scale factor $\lambda > 1$:

- **Constant returns to scale (CRS):** $f(\lambda L, \lambda K) = \lambda f(L, K)$.
- **Increasing returns to scale (IRS):** $f(\lambda L, \lambda K) > \lambda f(L, K)$.
- **Decreasing returns to scale (DRS):** $f(\lambda L, \lambda K) < \lambda f(L, K)$.

3.5.1 Returns to Scale for Cobb–Douglas

$Y = AL^\alpha K^\beta$. Then $f(\lambda L, \lambda K) = A(\lambda L)^\alpha (\lambda K)^\beta = \lambda^{\alpha+\beta} AL^\alpha K^\beta = \lambda^{\alpha+\beta} Y$.

Cobb–Douglas Returns

- $\alpha + \beta = 1$: **constant** returns to scale.
- $\alpha + \beta > 1$: **increasing** returns.
- $\alpha + \beta < 1$: **decreasing** returns.

Common Mistake

Returns-to-scale measures what happens when *all inputs scale up proportionally*. It is **different** from the marginal product of a single input (which holds others fixed). A Cobb–Douglas function can have constant returns to scale ($\alpha + \beta = 1$) while still exhibiting diminishing MP_L ($\alpha < 1$).

3.5.2 Linear and Leontief Returns

Both **linear** ($Y = aL + bK$) and **Leontief** ($Y = \min\{L/a, K/b\}$) production functions exhibit CRS by direct substitution.

3.5.3 Economies and Diseconomies of Scale (Cost Side)

- **Economies of scale** — LRAC *falls* as Q rises. Usually tied to IRS.
- **Diseconomies of scale** — LRAC *rises* as Q rises. Usually tied to DRS.
- **Constant economies of scale** — LRAC constant; CRS.

3.6 Short Run vs. Long Run

Definition

Long run (LR) — a period long enough that the firm can vary *all* inputs. **Short run (SR)** — a period in which *at least one* input (typically capital) is fixed.

SR and LR are not calendar concepts; they depend on the industry. “Two months” may be LR for a roadside vendor and SR for an aircraft manufacturer.

3.6.1 Why $\text{SRTC} \geq \text{LRTC}$

In the long run the firm optimises both L and K . In the short run with K stuck at $\bar{K}$, the firm can only adjust L . So:

$$\text{SRTC}(Q | \bar{K}) \geq \text{LRTC}(Q),$$

with equality only at the Q for which $\bar{K}$ is the long-run optimal capital level.

3.6.2 Short-Run Marginal Cost in Terms of MPL

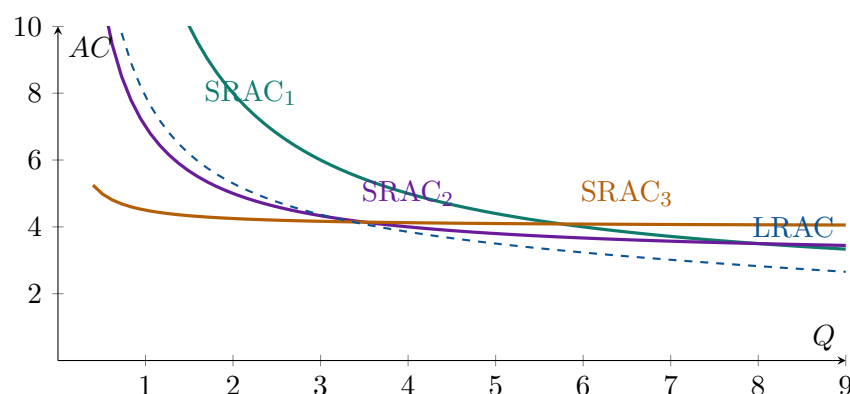
In the short run only labour varies. To produce one more unit of output, the firm needs $1/\text{MP}_L$ extra labour units, each costing w . Hence

$$\text{MC}_{\text{SR}} = \frac{w}{\text{MP}_L}$$

Diminishing $\text{MP}_L \Rightarrow$ rising MC_{SR} .

3.6.3 Envelope Property

The LRAC curve is the lower envelope of all possible SRAC curves (one for each value of fixed capital). At each Q , the LRAC touches the SRAC whose $\bar{K}$ is the long-run optimum for that Q .



3.7 From Cost Curves to Supply Function

Definition

A **supply function** $Q^S(P)$ gives the quantity a firm is willing to supply at each market price P . We derive it from cost minimisation + profit maximisation.

3.7.1 Step-by-Step Derivation (Price-Taker)

A price-taker firm faces P given. Profit maximisation: $P = MC(Q)$.

(i) **Use only the *upward-sloping* part of MC.** If MC is U-shaped, the equation $P = MC(Q)$ has two solutions. The downward-sloping solution is a local minimum of profit (intuitively: increasing Q slightly lowers MC below P and raises profit). Only the upward-sloping piece corresponds to a maximum.

(ii) **Stay above AVC — the shutdown rule.** If $P < AVC(Q^*)$, the firm cannot even cover its variable cost. It is better to *shut down* (produce $Q = 0$) than to operate at a loss greater than fixed cost.

Supply Function (Short-Run)

The (short-run) supply curve of a price-taking firm is

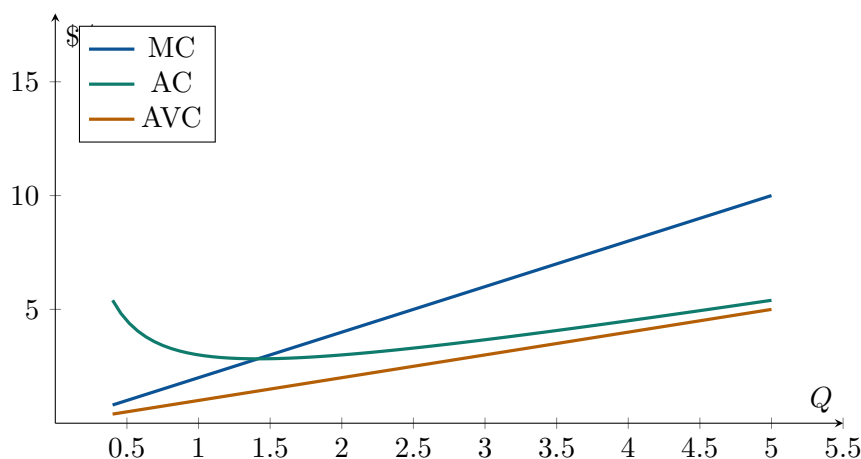
$$Q^S(P) = \begin{cases} \text{solution to } P = MC(Q) \text{ on rising branch} & \text{if } P \geq \min AVC, \\ 0 & \text{if } P < \min AVC. \end{cases}$$

The supply curve is the part of the MC curve that lies above the AVC curve.

3.7.2 Why Fixed Cost is Irrelevant to the Shutdown Decision

Fixed cost is sunk in the short run; you pay it whether you produce or not. So the *operating decision* compares revenue to *variable* cost.

3.7.3 Illustrative Diagram — Marginal Cost, AVC, AC, Supply



3.8 Detailed PYQ-Style Walkthrough — $TC = 200 + 50Q + Q^2$

Given. $TC(Q) = 200 + 50Q + Q^2$.

Identify:

$$TFC = 200, \quad TVC = 50Q + Q^2, \quad AVC = 50 + Q, \quad AFC = 200/Q,$$

$$AC = 50 + Q + 200/Q, \quad MC = 50 + 2Q.$$

Common PYQ subquestions.

- AFC at $Q = 100$: $200/100 = 2$.
- AVC at $Q = 25$: $50 + 25 = 75$.
- MC at $Q = 25$: $50 + 2 \cdot 25 = 100$.
- AC's minimum: solve $AC'(Q) = 0$ or use $MC = AC$.

3.9 End-of-Week Summary — Week 3**Week 3 Quick-Revision Sheet**

MRTS: $MRTS_{L,K} = MP_L / MP_K$ (magnitude of isoquant slope).

Cost Min: tangency $\Rightarrow MP_L / w = MP_K / r$ (equimarginal principle).

Recipes:

- Leontief: $L^* = aY_0$, $K^* = bY_0$, $C(Y_0) = (wa + rb)Y_0$.
- Linear: corner — use cheaper-per-output input.
- Cobb–Douglas $Y = L^\alpha K^\beta$: tangency $\Rightarrow K/L = (\beta w)/(\alpha r)$.

Returns to Scale (CD): $\alpha + \beta = 1$ CRS, > 1 IRS, < 1 DRS.

SR vs LR: $SRTC \geq LRAC$; LRAC is the lower envelope of SRACs.

$MC_{SR} = w / MP_L$.

Supply: upward-sloping part of MC, above min AVC. Below min AVC — shut down.

3.9.1 High-Probability Quiz Points (Week 3)

- Compute MRTS from a given $Y(L, K)$.
- Find L^*, K^* given w, r, Y_0 for linear / Leontief / Cobb–Douglas technology.
- Identify returns to scale.
- Recognise the shutdown rule.
- Compute MC in the short run as w / MP_L .

3.9.2 Practice Questions — Week 3

1. $Y = 4L^{1/2}K^{1/2}$, $w = 4$, $r = 16$, $Y_0 = 80$. Find L^*, K^*, C^* .
2. For $Y = 3L + 4K$, $w = 2$, $r = 5$. Determine the cost-minimising combination for $Y_0 = 60$.
3. State the returns to scale of $Y = 10L^{0.4}K^{0.3}$.
4. A firm in the SR has $w = 10$ and $MP_L = 4$. Compute MC.
5. Given $TC = 100 + 10Q$, what is the firm's supply curve, and at what P should it shut down?

4. Week 4 — Short-Run/Long-Run Costs and Consumer Theory

4.1 Recap and Roadmap

Week 4 closes the cost-curve story (short-run vs. long-run AC & MC) and then opens an entirely new chapter: **consumer behaviour**. We build, from first principles:

- Consumption sets, budget sets, budget lines.
- Preferences (rationality, continuity, monotonicity, convexity).
- Utility functions as a numerical representation of preferences.
- Indifference curves and the Marginal Rate of Substitution (MRS).

4.2 Short-Run vs. Long-Run Costs — Final Pieces

For the long run (constant returns to scale technology), the long-run total cost (LRTC) is linear, LRAC is constant, and LRMC = LRAC everywhere.

In the short run, capital is stuck at some $\bar{K}$. Let $\text{SRTC}(Q | \bar{K})$ be the resulting short-run total cost. The basic geometric facts:

Remember This

- SRTC has positive intercept $r\bar{K}$ on the cost axis = short-run fixed cost.
- $\text{SRTC} \geq \text{LRTC}$, with equality only at the Q for which $\bar{K}$ is long-run optimal.
- Higher $\bar{K} \Rightarrow$ higher fixed cost and (typically) higher MP_L , so $\text{MC}_{\text{SR}} = w/\text{MP}_L$ is *lower*.
- LRAC is the lower envelope of all SRAC curves.

4.2.1 Numerical Slope Insight

Recall $\text{MC}_{\text{SR}} = w/\text{MP}_L$. If $\bar{K}$ is large, capital complements labour (MP_L is higher), so the SRTC curve is *flatter*. If $\bar{K}$ is small, MP_L is lower and SRTC climbs more steeply. This is exactly what the geometry shows in lecture.

4.3 Building Blocks of Consumer Theory

The full picture, in order:

1. **Consumption Possibility Set** (everything imaginable).
2. **Consumption Feasibility Set** (what is actually attainable).
3. **Preferences** over bundles.
4. **Behavioural assumptions**.
5. **Utility function** as a numerical representation.

4.3.1 Consumption Bundles and Sets

A **bundle** is an n -tuple $x = (x_1, x_2, \dots, x_n)$ where x_i is the quantity of good i consumed. To make graphs feasible we restrict to $n = 2$ (e.g. “good 1” and “everything else”).

Consumption Possibility Set X . The set of all conceivable bundles. Restricted to the non-negative orthant: $X = \mathbb{R}_+^n = \{x : x_i \geq 0\}$. We allow *divisibility*: half a litre of milk is a valid bundle entry.

Consumption Feasibility Set Y (Budget Set). The bundles the consumer can *actually* afford given prices and income.

4.3.2 Budget Constraint — The Workhorse

With prices P_1, P_2 and income I , the consumer can afford bundle (x_1, x_2) if

$$P_1x_1 + P_2x_2 \leq I.$$

The **budget line** is the equality $P_1x_1 + P_2x_2 = I$:

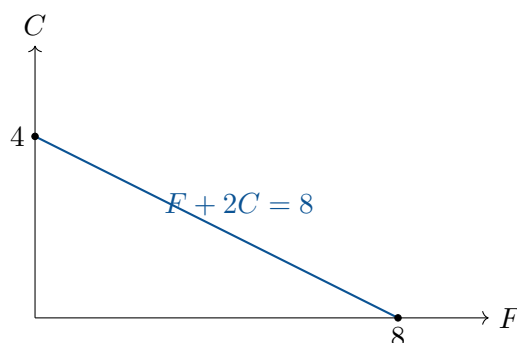
$$x_2 = \frac{I}{P_2} - \frac{P_1}{P_2}x_1.$$

Budget Line — Key Features

- **Vertical intercept** ($x_1 = 0$): $x_2 = I/P_2$ (all income on good 2).
- **Horizontal intercept** ($x_2 = 0$): $x_1 = I/P_1$ (all on good 1).
- **Slope**: $-P_1/P_2$ (the negative ratio of prices). Magnitude is the **exchange rate** of good 1 for good 2: give up one unit of good 1, you save P_1 rupees, which buys P_1/P_2 units of good 2.

4.3.3 Robinson Crusoe — Time as Income

Crusoe has 8 hours per day. 1 hour catches 1 unit of fish; 2 hours collects 1 unit of coconut. So the constraint $1 \cdot F + 2 \cdot C \leq 8$ is the budget line in *time-budget* form. Vertical intercept (only coconut): $C = 4$. Horizontal (only fish): $F = 8$. Slope: $-1/2$.



4.4 Shifts and Rotations of the Budget Line

Effects of Parameter Changes

- **Income I changes:** both intercepts change proportionally; slope unchanged. Parallel shift.
- **Price P_1 changes:** horizontal intercept I/P_1 changes; vertical intercept unchanged; slope changes (rotates around vertical intercept).
- **Price P_2 changes:** vertical intercept I/P_2 changes; horizontal unchanged; slope changes (rotates around horizontal intercept).

Worked example (Jan 2026 PYQ). Prices $P_C = 20$, $P_F = 10$, income $I = 1000$, with fish on the y-axis (so good 1 is coconut, good 2 is fish).

- Intercepts: x_1 -intercept (coconut only) = $1000/20 = 50$; x_2 -intercept (fish only) = $1000/10 = 100$.

- Slope of budget line (with fish on y): $-P_C/P_F = -20/10 = -2$.
- If P_C is *halved* to 10: new slope $= -10/10 = -1$; new intercepts: 100, 100.
- If income is halved (prices unchanged): slope same (-2) , both intercepts halved (25, 50).

4.5 Preferences — Rationality and Continuity

We use the symbol $\succeq$ for “at least as good as”. Read $x \succeq y$ as “bundle x is weakly preferred to y ”.

Rationality Axioms

- **Completeness:** For any x, y , either $x \succeq y$, $y \succeq x$, or both ($x \sim y$, indifferent).
- **Transitivity:** If $x \succeq y$ and $y \succeq z$, then $x \succeq z$.
- **Reflexivity:** $x \succeq x$.

A preference relation is **rational** if it satisfies all three.

Why do we assume rationality? Without it, we cannot consistently rank bundles, and no meaningful theory of choice is possible.

4.5.1 Continuity

Definition

Continuity: If $x_n \rightarrow x$ and $y_n \rightarrow y$ are sequences of bundles with $x_n \succeq y_n$ for all n , then $x \succeq y$. *Loosely:* no sudden jumps in preference; nearby bundles get nearby utilities.

4.5.2 The Existence of Utility Functions

Debreu's Theorem (Stated Informally)

If $\succeq$ is **rational and continuous** on the consumption set, then there exists a **continuous utility function** $u : X \rightarrow \mathbb{R}$ such that

$$x \succeq y \iff u(x) \geq u(y).$$

Utility is ordinal. Only the order matters — not the absolute number. Any strictly increasing transformation $g(u(x))$ (e.g. $u + 100$, $3u$, u^3 , $\ln u$) represents *exactly the same* preferences. We cannot say “Bundle A gives twice the happiness of Bundle B” — that statement has no meaning under ordinal utility.

4.6 Behavioural Assumptions on Preferences

4.6.1 (Local) Non-Satiation

Definition

Local non-satiation: For any bundle x and any $\varepsilon > 0$, there exists a bundle y with $\|y - x\| < \varepsilon$ such that $y \succ x$.

In plain English: in any tiny neighbourhood of any bundle, there is a strictly preferred bundle. *No bundle is a “bliss point.”*

4.6.2 Monotonicity (“More is Better”)

Definition

Monotonicity: If $y \geq x$ componentwise and $y \neq x$, then $y \succ x$.

Relationship between the two.

- Monotonicity $\Rightarrow$ Local non-satiation. (Just move slightly in the “more” direction.)
- Local non-satiation *does not imply* monotonicity. Example: “less is better” (strange but coherent). At the origin, no neighbourhood escapes the non-negative orthant — local non-satiation fails there but monotonicity (in the inverse sense) holds elsewhere.

4.6.3 Consequences for Indifference Curves

Remember This

Under monotonicity, indifference curves must be:

1. **Thin** (not regions). Two bundles cannot give the same utility if one has more of every good.
2. **Downward sloping**. An upward-sloping segment would mean a bundle with more of *both* goods has the same utility — contradiction.
3. **Non-intersecting** (different indifference curves cannot cross). Crossing implies a contradiction with transitivity.

4.7 Convexity of Preferences

Definition

Weak convexity: If $x \succeq y$, then for $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \succeq y$.

Strict convexity: The same holds with strict $\succ$ for $\lambda \in (0, 1)$.

Economic interpretation. *Balanced bundles are preferred to extreme bundles.* If you like both apples and bananas, a basket with some of each is at least as good as a basket with only one.

4.7.1 Convexity and the Shape of Indifference Curves

- **Convex preferences** $\Rightarrow$ indifference curves are **bowed toward the origin** (the upper-contour set is convex).
- **Linear** preferences (perfect substitutes) $\Rightarrow$ *straight* indifference curves (weakly convex but not strictly).
- **Concave preferences** $\Rightarrow$ “bowed out” indifference curves: extreme bundles preferred to mixes (unusual: e.g. “curd and chocolate, but not together”).

4.8 The Marginal Rate of Substitution (MRS)

Definition

The **Marginal Rate of Substitution** of good 1 for good 2 is the amount of good 2 the consumer must receive to compensate for the loss of one unit of good 1, keeping utility constant:

$$\text{MRS}_{1,2} = - \left. \frac{dx_2}{dx_1} \right|_{u=\text{const}} = \frac{MU_1}{MU_2},$$

where $MU_i = \partial u / \partial x_i$ is the marginal utility of good i .

Derivation. From $u(x_1, x_2) = \text{const}$, total differential:

$$\frac{\partial u}{\partial x_1} dx_1 + \frac{\partial u}{\partial x_2} dx_2 = 0 \Rightarrow \frac{dx_2}{dx_1} = - \frac{MU_1}{MU_2}.$$

MRTS vs. MRS — Twin Concepts

- **MRTS** (production) = MP_L / MP_K — slope of *isoquant*.
- **MRS** (consumption) = MU_1 / MU_2 — slope of *indifference curve*.

The mathematics is identical; the economic story is dual (production vs. consumption).

4.8.1 Diminishing MRS

Strict convexity of preferences $\Rightarrow$ diminishing MRS along an indifference curve. Intuition: as you consume more of good 1, you value it less on the margin (relative to good 2), and would give up less of good 2 to get more of it.

4.9 Canonical Utility Functions

Three forms parallel the three production functions of Weeks 2–3:

4.9.1 (1) Perfect Substitutes (Linear)

$$u(x_1, x_2) = ax_1 + bx_2.$$

MRS = a/b is constant; indifference curves are parallel straight lines.

Common Mistake

“Perfect substitutes” does *not* mean the exchange ratio is 1:1. It means MRS is **constant**. Rs 100 in dimes and Rs 100 in quarters can be perfect substitutes even though one quarter $\neq$ one dime.

4.9.2 (2) Perfect Complements (Leontief)

$$u(x_1, x_2) = \min\{ax_1, bx_2\}.$$

Indifference curves are L-shaped (right-angle kinks along the ray $bx_2 = ax_1$). Goods consumed only in fixed proportions (left shoe and right shoe; two quarters and a dime to operate a vending machine).

4.9.3 (3) Cobb–Douglas

$$u(x_1, x_2) = x_1^\alpha x_2^\beta.$$

Indifference curves are smooth convex curves; MRS = $\frac{\alpha x_2}{\beta x_1}$ (diminishing).

4.9.4 (4) “Bliss Point” / Quadratic

$$u(x_1, x_2) = -(x_1 - a)^2 - (x_2 - b)^2.$$

Indifference curves are concentric circles around the bliss point (a, b) . *More is not always better!* This violates monotonicity but illustrates a person with a “satiation” preference.

4.10 End-of-Week Summary — Week 4

Week 4 Quick-Revision Sheet

SR vs LR cost (continued):

- $\text{SRTC} \geq \text{LRTC}$, equality at the long-run optimal $\bar{K}$ for that Q .
- $\text{MC}_{\text{SR}} = w / \text{MP}_L$.
- LRAC is the lower envelope of SRAC curves.

Consumer Theory:

- Budget line: $P_1x_1 + P_2x_2 = I$; slope $-P_1/P_2$.
- Intercepts: $I/P_1, I/P_2$.
- Rationality (completeness, transitivity, reflexivity) + continuity $\Rightarrow$ existence of utility function.
- Utility is **ordinal**: invariant to monotone transformations.
- Monotonicity (more is better) $\Rightarrow$ ICs thin, downward-sloping, non-intersecting.
- Convexity $\Rightarrow$ ICs bowed toward origin; balanced bundles preferred.
- $\text{MRS} = \text{MU}_1 / \text{MU}_2$.

Canonical Utility Functions: linear (perfect substitutes), Leontief (perfect complements), Cobb–Douglas, bliss-point (concentric circles).

4.10.1 High-Probability Quiz Points (Week 4)

- Compute slope and intercepts of a budget line.
- Effect of changing P_1, P_2, I on the budget line.
- Identify perfect substitutes / complements from a description.
- Convex vs. non-convex preferences (Coke-machine PYQ: 2 quarters + 1 dime).
- Compute MRS from a utility function.

4.10.2 Practice Questions — Week 4

1. Budget line: $P_x = 5, P_y = 10, I = 100$. Draw it and label intercepts and slope.
2. If P_x doubles, what happens to the budget line?
3. For $u(x, y) = xy$, compute MRS at $(2, 8)$.
4. Are dimes and quarters convex preferences for Elmo (who needs exactly 2 quarters + 1 dime per drink)?
5. Find the MRS for $u = 3x + 2y$ (perfect substitutes).

Part II — Detailed PYQ Solutions

5. How to Read this Section

The PYQs cover **five terms**: 2024T1, 2024T2, 2025T1, 2025T2, and the Jan 2026 sample paper. Many questions repeat (Coke-machine, Fritos, Cobb–Douglas cost minimisation, TC quadratic). Rather than solving each duplicate, I group recurring problems into **question families**, solve the general case once, then catalogue every numerical variant.

For **every** question I provide:

1. **Interpretation** — what's being asked, which topic, the linguistic clues.
2. **Thought process** — how an experienced student approaches it.
3. **Solving strategy** — given, required, concept, formula, simplification.
4. **Formula explanation** — meaning, variables, assumptions, validity.
5. **Common mistakes, exam tricks**, and the **economic intuition** behind the answer.

6. Family 1 — Explicit, Implicit, and Sunk Costs (Snack Delivery + Fritos)

6.1 The Snack-Delivery Business

The Question (recurring in 2025T1 Q208–209, Jan 2026 Q22–23, 2025T2 Q257–258)

You start a snack delivery business on campus. You pay \$500/month for web-hosting and own an SUV with a \$300 monthly car payment and \$100/month insurance. Each order takes 0.5 hours and uses \$0.50 of gasoline. You pay grocers cash for goods. Your outside option is a \$6/hour campus dining-hall job (weekends only). Identify your explicit and implicit costs.

6.1.1 Q1 — “Your explicit costs include ...” (Answer: All of these)

Interpretation. Asks which of the listed items are *explicit* (out-of-pocket payments). Topic: cost classification.

Thought process. An explicit cost is *anything you write a cheque for*: web hosting, gasoline, payments to grocers. Each of those bullets is a cash outflow. Implicit costs (foregone wages, foregone interest) are *not paid out*.

Why each option:

- Leasing computer time — ✓ explicit (paid in cash).
- Gasoline — ✓ explicit (paid at the pump).
- Money to grocers — ✓ explicit.
- Therefore, “All of these” is correct.

Exam Trick

When an option “All of these” appears and *each individual item is uncontroversially* of the same type, the answer is almost always “All of these.”

6.1.2 Q2 — “Your implicit cost will be ...” (Answer: \$6/hour dining-hall wage)

Interpretation. Implicit cost is the value of the *next-best alternative* foregone. Topic: opportunity cost.

Thought process. The \$300 car payment and \$100 insurance are *paid in cash* $\Rightarrow$ explicit. The \$6/hour dining-hall wage is what you’d earn instead of operating the business — that’s an implicit cost.

Common Mistake. Many students pick “All of these,” forgetting that car payments and insurance are cash outflows already counted as explicit costs.

Common Mistake

Critical distinction: a *cost* can be either explicit or implicit, not both. The \$300 car payment is explicit. Forgoing \$6/hr at the dining hall is implicit.

6.2 The Fritos Family (Sunk Cost & Opportunity Cost)

Real-World Application

Generic story (with parameters P_0, f, S): You paid P_0 for 5 cases of Fritos the customer rejected. The grocer will refund you f rupees per rupee of returned merchandise (so total refund = $f \cdot P_0$). A fraternity offers S for the lot (self-pickup).

The five PYQ variants:

Term	P_0	f	S	Opportunity Cost
2024T1	\$100	0.25	\$55	\$25
2024T2	\$100	0.25	\$55	\$25
2025T1	\$100	0.25	\$55	\$25
2025T2	\$200	0.50	\$125	\$100
Jan 2026	\$200	0.50	\$125	\$100

6.2.1 General Solution

Interpretation. “Opportunity cost of filling this (fraternity) order” = value of the next-best alternative you give up to sell to the fraternity. Topic: opportunity cost + sunk cost.

Thought process. The original purchase price P_0 is **SUNK** — it does not depend on which future action you take. *Ignore it!* Once you have the cartons, your two actions are:

- Sell to the fraternity for S .
- Return to the grocer for $f \cdot P_0$.

The opportunity cost of selling to the fraternity is the value of the foregone alternative = $f \cdot P_0$.

Formula

$$\text{Opportunity cost (frat sale)} = f \cdot P_0, \quad \text{Net gain (frat sale)} = S - f \cdot P_0.$$

Sell if and only if $S > f \cdot P_0$.

Worked answers.

- 2024T1, T2 & 2025T1: $fP_0 = 0.25 \times 100 = 25$. Frat offer $\$55 > 25$. **Sell. OC = \$25.**
- 2025T2 & Jan 2026: $fP_0 = 0.5 \times 200 = 100$. Frat offer $\$125 > 100$. **Sell. OC = \$100.**
- **Sunk cost** is the original P_0 minus its recovered portion = $P_0(1 - f)$. For the 2025T1 variant: $\$100 \times 0.75 = \75 (this is asked as a sub-question).

Remember This

Two cardinal rules of this question family:

1. Ignore sunk costs.
2. Opportunity cost = next-best foregone, not the historical price.

7. Family 2 — Production-Function Identification

7.1 Q (2024T2/Jan 2026): “A firm can trade 3 units of labour for 1 unit of capital keeping output constant”

Topic. Identifying production functions from verbal cues. Topic: Week 2–3.

Thought process. “Always trade a units of one input for b of the other” means the exchange ratio (MRTS) is *constant* — the signature of **linear** (perfect substitute) technology, $Y = aL + bK$.

Why not the others.

- Cobb–Douglas: $MRTS = \frac{\alpha K}{\beta L}$ *varies* with L, K . $\times$
- Leontief: only the fixed proportion works; you cannot trade beyond the kink. $\times$

Answer: Linear production function.

7.2 Q (2024T2/Jan 2026): “Two full-time workers required per piece of machinery”

Thought process. “Exactly n workers per m machines” = **fixed-proportion** $\Rightarrow$ **Leontief**.

Answer: Leontief.

7.3 Q (2024T2/Jan 2026): “For $q = 4L^{1/2} + 4K \dots$ ”

Topic. Marginal products from a given production function.

Setup.

$$MP_L = \frac{\partial q}{\partial L} = 4 \cdot \frac{1}{2} L^{-1/2} = \frac{2}{\sqrt{L}}. \quad MP_K = \frac{\partial q}{\partial K} = 4.$$

Interpretation.

- $MP_L = 2/\sqrt{L}$ falls as L grows. *Diminishing*.
- $MP_K = 4$ is independent of K . *Constant*.

Answer: MPL decreasing, MPK constant.

7.4 Q (2024T1/2025T1): Cobb–Douglas Returns to Scale (general statement)

Question. “The Cobb–Douglas production function $F(K, L) = K^p L^q \dots$ ”

Answer. “Shows increasing, decreasing, or constant returns to scale based on the value of $p + q$.”

Reasoning. $F(\lambda K, \lambda L) = \lambda^{p+q} F(K, L)$. Hence:

$$\begin{cases} p + q = 1 : & \text{CRS} \\ p + q > 1 : & \text{IRS} \\ p + q < 1 : & \text{DRS} \end{cases}$$

Why the other options are wrong.

- “Only CRS” — false, depends on $p + q$.
- “Constant MP_L ” — false, $MP_L = qK^p L^{q-1}$ depends on L unless $q = 1$.
- “Constant MP_K ” — same logic.

7.5 Q (Jan 2026 only): Note on Cobb–Douglas variant

Potential Issue Detected

In the **Jan 2026 sample paper**, the Cobb–Douglas question reads “Production function $F(K, L) = K^p L^q$ ” and lists the correct option as “*May exhibit constant returns to scale.*” In the **2024T1** version the correct option is “*Shows increasing, decreasing or constant returns to scale based on $p + q$* ” — which is the *more accurate* statement.

Both options are technically true:

- “May exhibit CRS” is true (when $p + q = 1$) — but it is the *weakest* of the four options.
- “Depends on $p + q$ ” is the strongest, complete statement.

Recommendation. If both options appear on a paper, the **best answer** is the “depends on $p + q$ ” option. If only “may exhibit CRS” is available (as in Jan 2026), that becomes the correct option since the other three are unambiguously *wrong*. **Please verify the precise wording with the course instructor.**

8. Family 3 — Marginal Product Definitions and AP–MP Relations

8.1 Q: Definition of MP of Labour

Answer. “The extra output produced by employing one more unit of labour while holding other inputs constant.”

Why not others.

- “Total output / total labour” — that is *average* product, not marginal.
- “Holding other inputs vary” — that confuses partial with total derivative.
- “One more unit of capital” — that is MP_K , not MP_L .

8.2 Q: Definition of MP of Capital

Answer. “The extra output produced by employing one more unit of capital while holding other inputs constant.” (Jan 2026 Q5.)

8.3 Q: “If $MC = AC$, then AC is ...”

Answer. *Minimised.*

Reasoning. From Week 2:

$$\frac{d AC}{dQ} = \frac{1}{Q}(MC - AC).$$

When $MC = AC$, $d AC / dQ = 0$ — a stationary point. Since AC is U-shaped (falls then rises), the stationary point is a *minimum*.

8.4 Q: “When AP is decreasing in labour, $MP > AP$.”

Answer: False.

Reasoning. Mirror of MC – AC : $MP > AP \Rightarrow AP$ rising. If AP is decreasing, then $MP < AP$.

8.5 Q: “When AP is increasing, $MP < AP$.”

Answer: False. Same logic: AP rising $\Leftrightarrow MP > AP$.

8.6 Q: “When AP is at maximum, $MP = AP$.”

Answer: True. The MP curve cuts AP at its peak.

The MP–AP Rule (Memorise!)

$$\begin{cases} MP > AP & \Rightarrow AP \text{ rising} \\ MP < AP & \Rightarrow AP \text{ falling} \\ MP = AP & \Rightarrow AP \text{ at extremum (max here)} \end{cases}$$

Mnemonic: “The new score pulls the average toward it.” If your latest exam (MP) is above your current GPA (AP), your GPA rises.

9. Family 4 — Cobb–Douglas Cost Minimisation

9.1 The General Recipe

Master Recipe for Cobb–Douglas Cost Minimisation

Given $Q = AL^\alpha K^\beta$, input prices w (labour) and r (capital), and target Q_0 :

1. Tangency: $\frac{MP_L}{MP_K} = \frac{w}{r} \implies \frac{\alpha K}{\beta L} = \frac{w}{r}$.
2. Solve for K in terms of L : $K = \frac{\beta w}{\alpha r} L$.
3. Substitute into $Q_0 = AL^\alpha K^\beta$, solve for L^* , then K^* .
4. Cost $C^* = wL^* + rK^*$.

9.2 Q (2024T2 & Jan 2026): $q = 10L^{1/2}K^{1/2}$, $w = 20$, $r = 80$, $Q_0 = 100$.

Step 1 — MPs. $MP_L = 5K^{1/2}/L^{1/2}$, $MP_K = 5L^{1/2}/K^{1/2}$. $MRTS = K/L$.

Step 2 — Tangency. $K/L = w/r = 20/80 = 1/4 \Rightarrow K = L/4$.

Step 3 — Constraint. $100 = 10\sqrt{L \cdot L/4} = 10 \cdot L/2 = 5L \Rightarrow L^* = 20$.

Step 4 — K^* . $K^* = 20/4 = 5$.

Answer: $L^* = 20$, $K^* = 5$.

Verify the output: $q = 10\sqrt{20 \cdot 5} = 10\sqrt{100} = 100$. ✓

Total cost: $C^* = 20 \cdot 20 + 80 \cdot 5 = 400 + 400 = \800 .

9.3 Q (2025T1): $Q = 50\sqrt{LK}$, $w = 5$, $r = 20$, $Q_0 = 1000$.

MP_K / MP_L : For symmetric Cobb–Douglas, this equals L/K . So

$$\frac{MP_K}{MP_L} = \frac{L}{K} \stackrel{!}{=} ? \Leftrightarrow \text{We instead want } \frac{MP_L}{MP_K} = \frac{w}{r} = \frac{5}{20} = \frac{1}{4}.$$

So $MP_L / MP_K = K/L = 1/4 \Rightarrow K = L/4$.

The MCQ option “ratio $\partial Q / \partial L \div \partial Q / \partial K = ?$ ” was $1/4$. **Constraint.** $1000 = 50\sqrt{L \cdot L/4} = 25L \Rightarrow L^* = 40$, $K^* = 10$.

$L^* = 40$, $K^* = 10$.

9.4 Q (2025T2 — Same form, prices swapped): $Q = 50\sqrt{LK}$, $w = 20$, $r = 5$, $Q_0 = 1000$.

Tangency: $K/L = w/r = 20/5 = 4 \Rightarrow K = 4L$.

Constraint: $1000 = 50\sqrt{L \cdot 4L} = 100L \Rightarrow L^* = 10$, $K^* = 40$.

$L^* = 10$, $K^* = 40$.

Quiz Trap

Heads-up: The 2025T1 paper has $(w, r) = (5, 20)$ giving $(L^*, K^*) = (40, 10)$, while the 2025T2 paper has $(w, r) = (20, 5)$ giving $(L^*, K^*) = (10, 40)$. Mirror images! The cheaper input is used more intensively.

9.5 Q (2024T1): $q = 100KL$, $r = 1200$, $w = 300$, $q_0 = 10,000$.

MPs. $MP_L = 100K$, $MP_K = 100L$.

Tangency. $\frac{MP_L}{MP_K} = \frac{K}{L} = \frac{w}{r} = \frac{300}{1200} = \frac{1}{4} \Rightarrow K = L/4$, or equivalently $K/L = 0.25$ (asked as a sub-question).

Constraint. $10,000 = 100 \cdot K \cdot L = 100 \cdot (L/4) \cdot L = 25L^2$. $L^2 = 400 \Rightarrow L^* = 20$, $K^* = 5$.

Cost. $C^* = 300 \cdot 20 + 1200 \cdot 5 = 6000 + 6000 = \$12,000$.

$K/L = 0.25, L^* = 20, K^* = 5, C^* = \$12000.$

10. Family 5 — Chair Manufacturer (Perfect Substitute / Linear)

10.1 The Story

Generic Story

A chair manufacturer can produce one chair using 4 hours of **labour OR machinery** (in any combination). So $4L + 4K = \text{chairs}$, i.e. $q = L/4 + K/4$, equivalently $L + K = 4q$. Or think of it as: one chair takes 4 hours of *either* input. Wages w per hour of labour; rental r per hour of machinery. The firm currently uses 3 hours of one input per hour of the other.

This is the **Linear (perfect-substitute)** technology with $a = b = 1/4$, so $\text{MRTS} = 1$. Since labour and machinery are perfect substitutes at 1 : 1, the cheaper input dominates.

10.2 Variant: 2024T1 (Q235–237) $w = \$30$ labour, $r = \$15$ machinery.

Currently 3 hours of labour per hour of machinery. Is it cost-minimising? Compute productivity per dollar:

$$\frac{1}{w} = \frac{1}{30} \text{ (labour units of chair per \$)}, \quad \frac{1}{r} = \frac{1}{15} \text{ (machinery units per \$)}.$$

Machinery delivers *twice as much chair per dollar*, yet the firm is using *more* labour than machinery. **Not cost-minimising.**

Optimal combination. Use only the cheaper input (machinery). Each chair needs 4 hours of machinery: $L^* = 0$, $K^* = 4$.

$$L^* = 0, K^* = 4 \text{ (per chair).}$$

10.3 Variant: 2024T2 & Jan 2026 $w = \$15$ labour, $r = \$30$ machinery.

Prices reversed. Now labour is cheaper.

$$L^* = 4, K^* = 0 \text{ (per chair).}$$

Currently 3 hours of machine per hour of labour? Not cost-minimising — the firm uses too much of the *expensive* input.

The Universal Linear-Tech Rule

With perfect substitutes $Y = aL + bK$:

$$L^* > 0, K^* = 0 \iff \frac{\text{MP}_L}{w} > \frac{\text{MP}_K}{r} \iff \frac{a}{w} > \frac{b}{r}.$$

Otherwise the corner solution flips. (At equality, any mix works.)

11. Family 6 — Linear Production Function and Optimal Inputs

11.1 Q (2025T1): $Q = 10L + 2K$, $w = 5$, $r = 2$, $Q_0 = 200$.

MPs. $MP_L = 10$, $MP_K = 2$.

Per-dollar productivity. $MP_L/w = 10/5 = 2$ vs. $MP_K/r = 2/2 = 1$. Labour wins. Use only labour: $L^* = 200/10 = 20$, $K^* = 0$. $TC = 5 \cdot 20 = \$100$.

$L^* = 20$, $K^* = 0$, $TC = \$100$.

11.2 Q (2025T2): $Q = 5L + 2K$, $w = 5$, $r = 2$, $Q_0 = 200$.

Per-dollar. $MP_L/w = 5/5 = 1$ vs. $MP_K/r = 2/2 = 1$. **Tie!** Any combination satisfying $5L + 2K = 200$ minimises cost.

The PYQ answer key says $L^* = 0$, $K^* = 100$, $TC = \$200$. Let's verify: $5 \cdot 0 + 2 \cdot 100 = 200$ output. $TC = 5 \cdot 0 + 2 \cdot 100 = \200 . Alternative $L^* = 40$, $K^* = 0$: $TC = 5 \cdot 40 = \$200$. Same total cost!

Multiple Valid Answers

In the 2025T2 variant ($Q = 5L + 2K$, $w = 5$, $r = 2$), the answer key uses $K^* = 100$, $L^* = 0$, but *any* (L^*, K^*) satisfying $5L^* + 2K^* = 200$ minimises cost (the tie case). The TC is uniquely \$200. **Please verify with the course instructor whether the software accepts (20, 50), (40, 0), etc. as equally correct.**

12. Family 7 — The Coke-Machine (Leontief / Perfect Complement)

12.1 Q (2024T1, 2025T1)

Story. Coke machine requires *exactly* 2 quarters + 1 dime per drink. Elmo has 4 quarters and 2 dimes.

12.1.1 Sub-Q: How many drinks?

Topic. Leontief utility / perfect complements.

Setup. Let q = quarters, d = dimes. One drink needs 2 quarters AND 1 dime.

$$\# \text{ drinks} = \min\left\{\frac{q}{2}, \frac{d}{1}\right\} = \min\{4/2, 2/1\} = \min\{2, 2\} = 2.$$

2 drinks.

12.1.2 Sub-Q: Are Elmo's preferences convex?

Answer: Yes.

Reasoning. Leontief preferences have weakly convex upper-contour sets. The indifference curve is L-shaped; the upper-contour set (north-east of the kink) is a convex region. Hence preferences are convex (weakly).

Common mistake. Students confuse “goods don’t substitute” with “not convex.” But non-substitutability is consistent with convexity; the upper-contour set is still convex.

13. Family 8 — Budget-Line Comprehension (Robin/Coconut-Fish)

13.1 The Generic Setup

Two goods (coconut C and fish F), prices $P_C = 20$, $P_F = 10$, income $I = 1000$. Fish on the Y-axis. (Repeated in 2024T2, Jan 2026.)

Master Budget-Line Formulae

$$P_C \cdot C + P_F \cdot F = I, \quad F = \frac{I}{P_F} - \frac{P_C}{P_F} C.$$

- Slope $\frac{dF}{dC} = -\frac{P_C}{P_F}$.
- X-intercept (only coconut) $= I/P_C$.
- Y-intercept (only fish) $= I/P_F$.

Q: Slope of the budget line. $-P_C/P_F = -20/10 = -2$. **Answer:** -2 .

Q: Intercepts. X-intercept $= 1000/20 = 50$ (coconut); Y-intercept $= 1000/10 = 100$ (fish). **Answer:** 50 on X, 100 on Y.

Q: If P_C is halved to 10. New slope $= -10/10 = -1$. **Answer:** -1 .

Q: New intercepts. X-intercept $= 1000/10 = 100$; Y-intercept unchanged $= 100$. **Answer:** 100 on X, 100 on Y.

Q: Income halved, prices unchanged. Slope $= -P_C/P_F$ unchanged. Both intercepts halved (from 50, 100 to 25, 50). **Answer:** Slope same, both intercepts halved.

Visual Mnemonic for Budget-Line Shifts

- Income up $\Rightarrow$ Parallel shift outward.
- P_x up $\Rightarrow$ x-intercept moves in; line rotates around y-intercept (steeper if P_x alone changes).
- P_y up $\Rightarrow$ y-intercept moves in; line rotates around x-intercept (flatter if P_y alone increases).

14. Family 9 — $TC = a + bq + cq^2$ (Quadratic Short-Run Cost)

14.1 Master Recipe

From $TC = a + bq + cq^2$, Read Off Everything

$$TFC = a, \quad TVC = bq + cq^2, \quad AFC = \frac{a}{q}, \quad AVC = b + cq, \quad AC = \frac{a}{q} + b + cq, \quad MC = b + 2cq.$$

14.2 Q (2024T1): $TC = 100 + 25q + q^2$, $q = 100$ (or as asked).

- AFC at $q = 100$: $100/100 = 1$. **Answer: 1.**
- MC at $q = 25$: $25 + 2 \cdot 25 = 75$. **Answer: 75.**
- AVC at $q = 25$: $25 + 25 = 50$. **Answer: 50.**

14.3 Q (2025T1): $TC = 200 + 50q + q^2$, $q = 100$ or 25.

- AFC at $q = 100$: $200/100 = 2$. **Answer: 2.**
- MC at $q = 25$: $50 + 50 = 100$. **Answer: 100.**
- AVC at $q = 25$: $50 + 25 = 75$. **Answer: 75.**

14.4 Q (2025T2): $TC = 300 + 50q + q^2$, $q = 50$.

- AFC at $q = 50$: $300/50 = 6$. **Answer: 6.**
- MC at $q = 50$: $50 + 2 \cdot 50 = 150$. **Answer: 150.**
- AVC at $q = 50$: $50 + 50 = 100$. **Answer: 100.**

15. Family 10 — Linear TC: $TC = 200 + 55q$

PYQ (Jan 2026, 2024T2). $TC = 200 + 55q$ is linear — $TVC = 55q$, $AVC = 55$ (constant).

- Fixed cost: **200**.
- AVC at $q = 100$: $55q/q = 55$. **Answer: 55.**

16. Family 11 — MRTS, MRS, and Marginal Product Numerics

16.1 Q (Jan 2026 Q19): MP of capital from MRTS and MP_L .

Question. MP of labour in chip production is 50 kg/hour. MRTS of labour for machine capital is $1/4$. Find MP_K .

Concept.

$$MRTS_{L,K} = \frac{MP_L}{MP_K} \implies MP_K = \frac{MP_L}{MRTS_{L,K}}.$$

Compute. $MP_K = 50/(1/4) = 200$ kg/hour.

$MP_K = 200$ kg/hour.

Intuition. If giving up 1 hour of capital frees up 4 hours of labour while keeping output constant, capital is 4 times as productive on the margin — so $MP_K = 4 \cdot MP_L = 200$.

17. Family 12 — Diminishing Marginal Utility (2025T2 Q261–262)

17.1 Story.

$$U(x, y) = y\sqrt{x}.$$

17.1.1 Q261 — “Correct alternative”

Compute marginal utilities.

$$MU_x = \frac{\partial U}{\partial x} = \frac{y}{2\sqrt{x}}, \quad MU_y = \frac{\partial U}{\partial y} = \sqrt{x}.$$

Both are positive for $x, y > 0$, so **more is better** for both goods (monotonicity holds).

Is MU_x independent of x ? No — it has $\sqrt{x}$ in the denominator. Is MU_y independent of y ? Yes — $MU_y = \sqrt{x}$ doesn't depend on y .

The option “MU of x independent of x AND MU of y independent of y ” is partly true but the first half is *false*, so the whole option fails. The **only fully correct option** is “**more is better for each good**”.

Answer: more is better for each good.

17.1.2 Q262 — “Incorrect alternative”

Compute second derivatives.

$$\frac{\partial MU_x}{\partial x} = -\frac{y}{4x^{3/2}} < 0, \quad \frac{\partial MU_y}{\partial y} = 0.$$

MU_x diminishes in x . MU_y does *not* diminish in y (it's constant in y).

So: “Preferences exhibit diminishing MU of x ” ✓ true. “Diminishing MU of y ” ✗ false. The *incorrect* statement is the one about y .

Answer: “Diminishing marginal utility of y ” is incorrect.

18. Family 13 — Preference Axiom Violations (2024T1)

18.1 Q: “Cannot compare two goods — preference violates ...”

Answer: Completeness. Completeness says any two bundles can be compared. Inability to compare violates this axiom.

18.2 Q: Mango $\succ$ Apple, Apple $\succ$ Guava, Guava $\succ$ Mango. Violates ...

Answer: Transitivity.

Reasoning. The preferences form a *cycle*: $M > A > G > M$. Transitivity demands that if $M > A$ and $A > G$, then $M > G$ — but we are told $G > M$. Contradiction.

Common Trap

Don't confuse *transitivity violations* (cyclical) with *incompleteness* (“unable to compare”). Transitivity requires a *contradiction over three* bundles.

19. Family 14 — Gradient Mini-Questions (2025T1 Q205–206)

These appeared as preliminary calculus tune-up. Given $a = 50$, $b = 2$, with a function whose gradients $\partial/\partial a$ and $\partial/\partial b$ evaluate to 3 and 55 respectively. **Possible answers: 3 and 55.** These are likely from a Cobb–Douglas style expression evaluated at specific values; without the original LaTeX expression we record the answers as listed in the key.

OCR Limitation

The question stems in the 2025T1 paper Q205–206 are partially OCR-garbled (“ $a = 50, b = 2$, $\partial e/\partial a = ?$ ”). **Please verify the original expression with your professor.** The numerical answers (3 and 55) are taken directly from the answer key.

20. Family 15 — Game Theory: Entry Deterrence (2024T2)

Note. Game theory was introduced in the 2024T2 quiz as a stretch topic. It does not appear in the standard Weeks 1–4 syllabus, so this family is also covered in greater depth in **Part III**. Here we focus on the specific PYQ.

20.1 Q162–Q164: Entry deterrence sequential game

Setup (paraphrased from PYQ). A firm (Player 1) considers entering an established industry with one incumbent firm (Player 2). Player 1 chooses **Enter (E)** or **Not Enter (N)**. If Player 1 enters, Player 2 chooses **Accommodate (A)** or **Fight (F)**. Preferences:

- **Player 1:** $(E, A) \succ (N, \cdot) \succ (E, F)$. Most prefers entering unopposed; least prefers entering into a price war.
- **Player 2:** $(N, \cdot) \succ (E, A) \succ (E, F)$. Most prefers no entry at all; least prefers fighting an entrant.

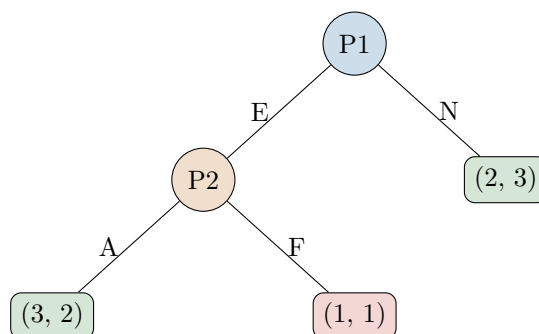
20.1.1 Step 1 — Assign ordinal payoffs

Ordinal means *order matters, magnitudes don't*. Use $3 > 2 > 1$:

$$\text{Player 1: } (E, A) = 3, \quad (N, \cdot) = 2, \quad (E, F) = 1.$$

$$\text{Player 2: } (N, \cdot) = 3, \quad (E, A) = 2, \quad (E, F) = 1.$$

20.1.2 Step 2 — Draw the extensive-form game tree



20.1.3 Step 3 — Number of subgames

Subgame — Definition

A **subgame** is any portion of the game tree that (i) starts at a single decision node, (ii) contains all subsequent nodes and branches from that node, and (iii) does not break any information sets.

In this game:

1. The **whole game** starting at Player 1's node — 1 subgame.
2. The **subgame** starting at Player 2's node (after E) — 1 subgame.

Number of subgames = 2.

20.1.4 Step 4 — Subgame perfect equilibrium via backward induction

Backward induction. Solve the last (deepest) decision first; substitute its outcome; then solve the next-to-last; and so on.

- **Player 2's decision (after E):** compare A (payoff 2) vs F (payoff 1). Choose A .
- **Player 1's decision (root):** given P2 will play A , compare E (which now yields payoff 3) vs N (payoff 2). Choose E .

SPE = (E, A) : Player 1 enters, Player 2 accommodates.

Intuition — “credible threats”

The incumbent *would like* to deter entry by threatening to fight. But that threat is **not credible** — if entry actually happens, fighting hurts the incumbent worse than accommodating ($1 < 2$). Knowing this, the entrant ignores the threat and enters. SPE eliminates non-credible threats.

21. Family 16 — Socio-Economic Classification (SEC) (2025T2 Q254–255)

Note. These questions test the Indian SEC framework, a marketing/economics segmentation grid used to categorize households by the chief wage earner's education and occupation. Out of syllabus for the lecture material but appears in the PYQ. Brief coverage here; deeper coverage in Part III.

21.1 The SEC grid (urban India, NCCS 2011)

Education ↓ / Occ. →	Unskilled	Skilled	Petty	Shop	Supervisor	Officer Jr.	Officer Sr.
Illiterate	E3	E2	D	D	D	D	D
School < 5	E2	E1	D	D	D	D	D
School 5–9	E1	D	C	C	B2	B2	A2
SSC/HSC	D	C	B2	B2	B1	B1	A2
Some college	D	B2	B1	B1	A2	A2	A1
Graduate/PG general	D	B1	A2	A2	A2	A1	A1
Graduate/PG prof.	D	B1	A2	A1	A1	A1	A1

Reading the grid. Top-left = lowest socio-economic status (E3); bottom-right = highest (A1). The classification depends on *both* the education and the occupation of the chief wage earner.

21.2 Q254 — “Family with three children, husband-wife, [chief wage earner profile X]”

The answer key marks **A2**. Without the full OCR'd description of the chief wage earner's education/occupation (the page was truncated), we trust the key.

21.3 Q255 — Husband: 12th-pass bodyguard; wife: literate cook

Reasoning.

- Chief wage earner: husband.
- Occupation: bodyguard → *skilled worker* / *personal service* category.
- Education: SSC/HSC (passed 12th).

Cross-referencing the grid: SSC/HSC row × Skilled column → **C**.

Answer: C.

Exam Tip

For SEC questions, identify (1) who the chief wage earner is, (2) their highest education level, and (3) their occupation category. Then read off the grid. Don't be misled by the spouse's details unless they are the chief earner.

Part III — Concepts Found Only in PYQs

Part III — Extra Concepts

About Part III

This part covers concepts that appeared in the PYQs but are **not in the formal Weeks 1–4 syllabus**. They are useful as background but may or may not be on your specific quiz. Read them once for context.

22. Game Theory — Sequential Games and SPE

22.1 Why game theory?

Standard managerial economics models a firm as if it chose quantity in isolation. But in many markets, a firm's best move depends on what *other firms* will do — and they in turn anticipate yours. **Game theory** is the formal study of such strategic interaction.

22.2 Building blocks

Key terms

- **Players:** the decision-makers.
- **Strategies:** the complete plan of action for each contingency.
- **Payoffs:** the numerical reward each player receives at the end.
- **Extensive form:** a tree diagram with nodes (decisions) and branches (choices), payoffs at the leaves.
- **Normal (strategic) form:** a payoff matrix — used when moves are simultaneous.

22.3 Ordinal vs Cardinal payoffs

- **Cardinal:** actual numbers matter. Payoff 10 is “twice as good” as 5.
- **Ordinal:** only the *ranking* matters. Any monotonic relabelling ($1 < 2 < 3$ or $100 < 200 < 700$) gives the same game.

PYQ-style problems usually ask you to assign *any* ordinal payoffs that respect the stated preferences. Use 1, 2, 3, ... for simplicity.

22.4 Nash Equilibrium

Nash Equilibrium

A profile of strategies $(s_1^*, s_2^*, \dots)$ such that *no player can do strictly better by unilaterally changing their own strategy*, holding all others fixed.

Intuition. At a Nash equilibrium, everyone's strategy is a best response to everyone else's. No regret.

22.5 Subgame Perfect Equilibrium (SPE)

Nash equilibrium has a weakness in sequential games — it admits *non-credible threats*. SPE strengthens it: a strategy profile is SPE if it is a Nash equilibrium *in every subgame*.

How to find SPE: backward induction. Start at the deepest node, find the player's best move there, replace that subtree by the resulting payoff vector, and recurse toward the root.

22.6 Counting subgames

A **proper subgame** starts at any *singleton decision node* (a single decision node not in an information set with another node) and contains everything that follows. In a tree with no information sets:

Memory Trick

Number of subgames = (number of decision nodes) for trees with perfect information and no information sets.

For the 2024T2 entry game, decision nodes = 2 (P1 root, P2 child) → 2 subgames.

23. Socio-Economic Classification (SEC / NCCS)

23.1 Origins

The SEC framework was developed by the Indian Market Research Society (IMRS) and is maintained today as **NCCS (New Consumer Classification System)** by the Media Research Users Council (MRUC). It is the de-facto standard for segmenting Indian households by socio-economic status.

23.2 What it classifies

NCCS classifies a household (urban) into one of **12 grades**: A1, A2, B1, B2, C, D, E1, E2, E3, plus rural variants. Classification is by the **Chief Wage Earner's** education × occupation grid.

23.3 Why managers care

- **Pricing strategy:** A1/A2 households tolerate premium pricing.
- **Channel selection:** D/E households are reached via mass media, not digital.
- **Product mix:** sachet-sized packaging targets C/D/E; large-format SKUs target A.
- **Media planning:** ad spend allocation depends on the target SEC mix.

23.4 Common mistakes

Don't confuse with

Income alone. SEC is not just income — a school teacher (A2) may earn less than an unskilled construction worker overtime (E1). Education weights heavily.

Appendices

Appendix A — Master Formula Sheet

Costs

$$\begin{aligned}
 TC &= TFC + TVC \\
 AC &= \frac{TC}{Q} \quad AFC = \frac{TFC}{Q} \quad AVC = \frac{TVC}{Q} \\
 MC &= \frac{dTC}{dQ} = \frac{dTVC}{dQ} \\
 \text{For } TC &= a + bq + cq^2 : \quad MC = b + 2cq, \quad AC = \frac{a}{q} + b + cq, \quad AVC = b + cq \\
 AC \text{ minimised at: } &MC = AC \\
 \text{Shutdown: } &P < \min AVC \\
 \text{Breakeven: } &P = \min AC
 \end{aligned}$$

Production

$$\begin{aligned}
 AP_L &= Q/L, \quad MP_L = \partial Q / \partial L, \quad MP_K = \partial Q / \partial K \\
 MRTS_{LK} &= \frac{MP_L}{MP_K} = - \frac{dK}{dL} \Big|_{Q=\bar{Q}} \\
 \text{Cost-min tangency: } &\frac{MP_L}{w} = \frac{MP_K}{r} \\
 \text{Isocost: } &wL + rK = C, \quad \text{slope} = -w/r
 \end{aligned}$$

Profit Maximisation

$$\begin{aligned}
 \pi(Q) &= TR(Q) - TC(Q) \\
 \text{FOC: } &MR = MC, \quad \text{SOC: } \frac{dMC}{dQ} > \frac{dMR}{dQ}
 \end{aligned}$$

Consumer Theory

$$\begin{aligned}
 \text{Budget: } &P_1x_1 + P_2x_2 = I, \quad \text{slope} = -P_1/P_2 \\
 MU_i &= \partial U / \partial x_i \\
 MRS_{12} &= \frac{MU_1}{MU_2} = - \frac{dx_2}{dx_1} \Big|_{U=\bar{U}} \\
 \text{Optimum: } &MRS_{12} = P_1/P_2
 \end{aligned}$$

Canonical Functional Forms

Form	Production	Utility
Linear/Perfect substitutes	$Q = aL + bK$	$U = ax + by$
Leontief/Perfect complements	$Q = \min\{aL, bK\}$	$U = \min\{ax, by\}$
Cobb–Douglas	$Q = AL^\alpha K^\beta$	$U = x^\alpha y^\beta$

Appendix B — Glossary of Key Terms

Accounting profit Total revenue minus *explicit* costs.

Average cost (AC) Total cost per unit of output, TC/Q .

Budget line The boundary of the budget set; combinations of two goods costing exactly the consumer's income.

Choke price The lowest price at which quantity demanded is zero.

Cobb–Douglas Functional form $Q = AL^\alpha K^\beta$; widely used because it has smooth tangencies and constant cost-share parameters.

Comparative statics How an equilibrium changes when an exogenous parameter changes.

Consumer surplus Area between the demand curve and the price line; benefit to consumers from being able to buy at a given price.

Demand curve Quantity demanded as a function of price (downward-sloping under the law of demand).

Economic profit Total revenue minus *all* costs, including implicit (opportunity) costs.

Economies of scale Falling LRAC as output expands.

Envelope LRAC is the lower envelope of all SRAC curves — it touches each from below.

Equimarginal principle At cost-min: $MP_L/w = MP_K/r$. At utility-max: $MU_1/P_1 = MU_2/P_2$.

Explicit cost Out-of-pocket monetary payment.

Fixed cost Cost that doesn't change with output in the short run.

Implicit cost Opportunity cost of resources owned (foregone wages, foregone rent).

Indifference curve Set of consumption bundles giving equal utility.

Isocost line Set of input bundles costing the same total expenditure.

Isoquant Set of input combinations producing the same output.

Leontief Functional form $\min\{aL, bK\}$ — perfect complements; no substitution.

Long run Time horizon in which all inputs (including capital) are variable.

Marginal cost (MC) Cost of producing one more unit, dTC/dQ .

Marginal product Extra output from one more unit of input, $\partial Q/\partial L$.

Marginal revenue Extra revenue from one more unit sold, dTR/dQ .

Marginal utility Extra utility from one more unit of consumption.

MRS Marginal rate of substitution; ratio of marginal utilities; slope magnitude of the indifference curve.

MRTS Marginal rate of technical substitution; ratio of marginal products; slope magnitude of the isoquant.

Nash equilibrium Strategy profile where no player wants to unilaterally deviate.

Opportunity cost Value of the best alternative foregone.

Production function Mapping from inputs to maximum producible output.

Quasi-fixed cost Zero if not producing; positive flat amount once producing.

Rational preferences Complete, transitive, reflexive.

Returns to scale How output responds when all inputs scale by the same factor.

Reservation price Maximum a buyer is willing to pay, or minimum a seller is willing to accept.

Short run Horizon in which at least one input (typically capital) is fixed.

Shutdown price $\min AVC$; below this, the firm should produce zero in the short run.

SPE Subgame perfect equilibrium; Nash equilibrium in every subgame.

Sunk cost An already-incurred, unrecoverable cost; irrelevant for forward-looking decisions.

Total revenue (TR) Price times quantity, $P \cdot Q$.

Utility function Numerical representation of preferences.

Variable cost Cost that varies with output.

Appendix C — Quick Revision: One-Pager

The Whole Course in One Box

Week 1. Economics = allocating scarce resources. Managerial economics applies microeconomic tools to business decisions. Demand curve aggregates reservation prices; consumer surplus = area between demand and price; choke price = lowest P at which $Q^d = 0$. Profit = TR – TC. Market structures range from perfect competition to monopoly.

Week 2. Costs: explicit + implicit; sunk costs are ignored. Accounting profit $\neq$ economic profit. $TC = TFC + TVC$. $MC = dTVC/dQ$. MC cuts AC at its minimum. Production functions: linear, Leontief, Cobb–Douglas. MP pulls AP towards it. Profit-max: $MR = MC$.

Week 3. $MRTS = MP_L/MP_K$. Cost-min: tangency at $MP_L/w = MP_K/r$. Cost function inherits returns-to-scale from production. $SR \geq LR$ cost. LRAC is the envelope of SRAC. Supply = MC above $\min AVC$.

Week 4. Consumer theory: rational preferences (complete, transitive, reflexive) + continuity $\Rightarrow$ utility function exists (Debreu). $MRS = MU_1/MU_2$. Budget line $P_1x_1 + P_2x_2 = I$. Canonical utility functions match canonical production functions.

Top 10 high-probability quiz topics

1. Cobb–Douglas cost minimisation: $L^* = (\alpha/\beta)(r/w)K^*$.
2. Quadratic TC $\rightarrow$ MC, AVC, AC, breakeven, shutdown.
3. Explicit vs implicit cost identification (Fritos/snack-shop variants).
4. Linear technology corner solution (chair-manufacturer).
5. Budget-line slope and shifts (Coconut–Fish).
6. MP/AP from a Cobb–Douglas production function.
7. Identify the production function from a verbal description (perfect subs, Leontief, CD).
8. MRS from utility, MRTS from production.
9. Preference axiom violations (cyclical $\rightarrow$ transitivity; incomparable $\rightarrow$ completeness).
10. Returns to scale check: $\alpha + \beta = 1 / > 1 / < 1$.

Closing Note

A note to the reader

This e-book grew out of lecture transcripts, slides, and five quizzes for the IIT Madras BS Data Science Managerial Economics course (Prof. Vimal Kumar). It is designed as a **living document**: as new weeks or PYQs are released, new chapters can be appended to the same source file.

If you spot an error, an unclear explanation, or a place where a diagram could be sharper, make a note — and update accordingly. Good notes are never finished; they are merely abandoned in time for the exam.

— *Made by Anmol Kansal with AI*